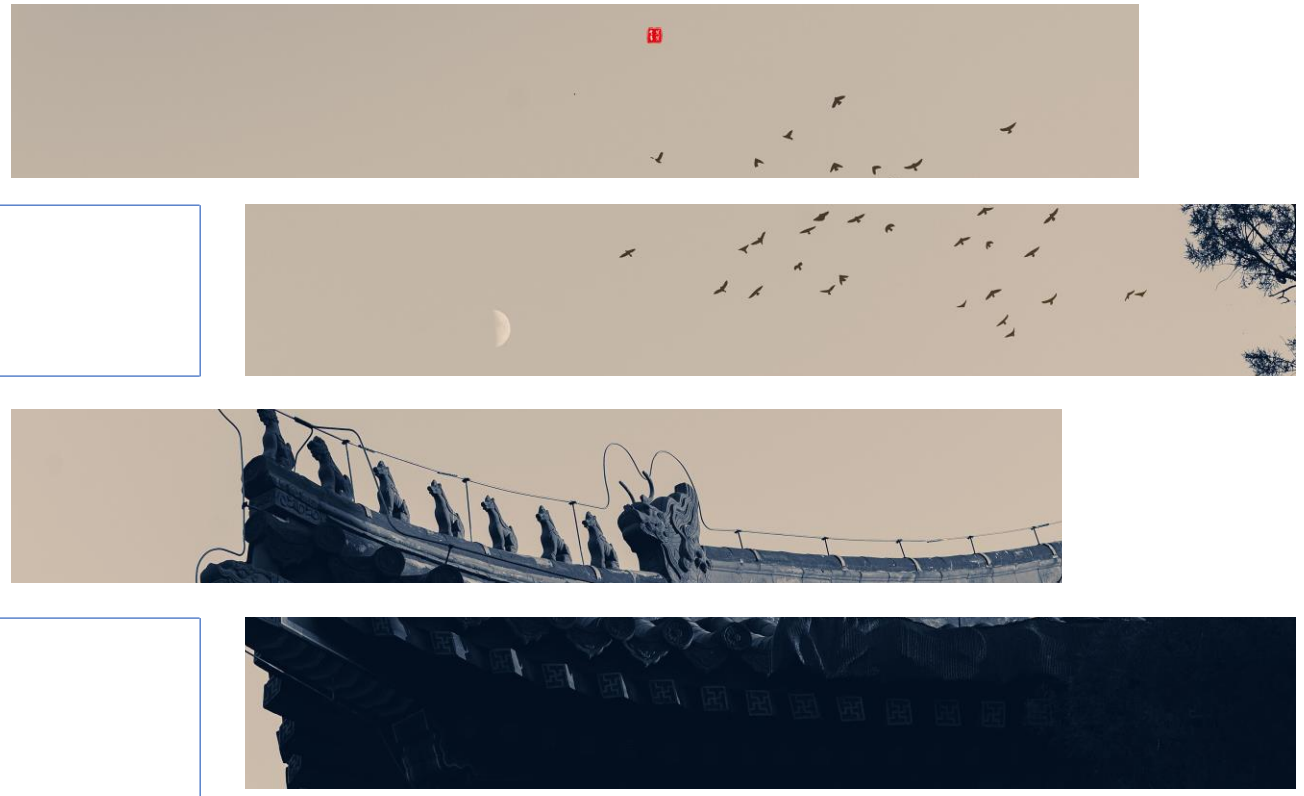
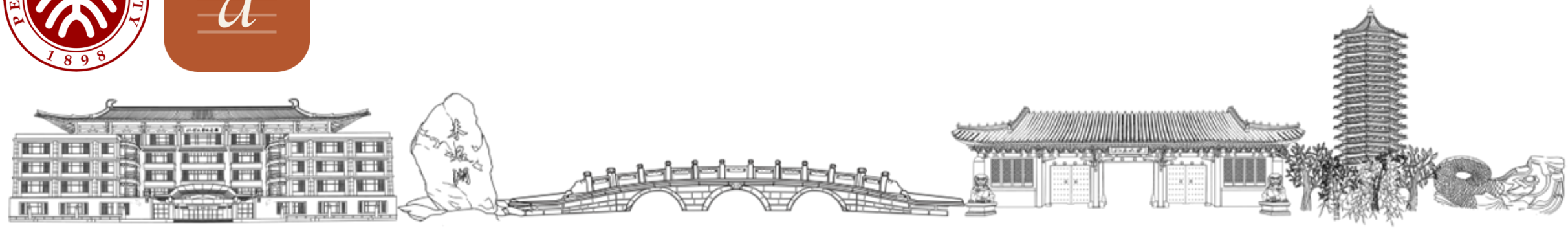
Cognitive Reasoning
Fall 2026



Lecture 03
Affordance, Functionality, and HOIs

Instructor: Dr. Chi Zhang
School of Intelligence Science and Technology



Logistics

- Group: 10.7
- Proposal check: ~10.23

What do you see in this image?



What do you see in this image?



7:43 PM

Claude-Sonnet-4-Reasoning

Thinking...

> I can see what appears to be an official or government building interior, with several people in business suits. There's a security checkpoint with metal detectors that people are going through. The setting looks formal and official, with multiple men in dark suits. The flooring appears to be a checkered pattern, and there are white walls with some displays or screens visible. This looks like it could be at a government building, airport, or other secure facility where people need to go through security screening.
>
> The people are dressed formally in business attire, suggesting this is likely some kind of official or diplomatic event. There appear to be security personnel and what might be officials or dignitaries going through a security checkpoint process.

This image shows what appears to be a security checkpoint in an official building. Several men in dark business suits are going through metal detectors, with one person in the center appearing to walk through the security scanner while others wait in line behind him. The setting looks formal and institutional, with white walls, a checkered floor pattern, and what appears to be displays or information boards on the walls. The scene suggests this is likely taking place in a government building, embassy, or similar secure facility where visitors must pass through security screening. The formal attire of all the individuals indicates this is probably an official or diplomatic event.

7:43 PM

@GPT-5

7:45 PM

GPT-5

It looks like a lighthearted moment in a locker-room or gym-like hallway with green-and-white checkered tile. Several men in business suits are gathered, smiling and laughing. One man is standing on a mechanical scale near a mirror; another man behind him is playfully pressing a foot down on the back of the scale platform, as if to skew the reading. Others in the background appear amused, holding folders and papers. The setting has lockers, mirrors, and countertops along the walls, not a security checkpoint.

7:45 PM

@Grok-4

7:46 PM

Grok-4

This image captures a humorous scene in what appears to be a school or gym locker room. A group of men, all dressed in formal business suits, are gathered around a mechanical bathroom scale mounted on the wall near a mirror. The central figure—a tall man in a gray suit—is standing on the scale, holding a clipboard or folder, while looking down at it. Directly behind him, another man in a dark suit is playfully placing his foot on the back edge of the scale platform, seemingly to manipulate the weight reading and make it higher. The group, including others in the background holding documents and smiling or laughing, seems to be enjoying the lighthearted prank. The setting features wooden lockers along the walls, mirrors, sinks or countertops, and a green-and-white checkered tile floor, evoking a casual, behind-the-scenes moment among professionals.

7:46 PM

If you were tasked to build a model …

1. What do you need to **parse this image**?

- Object detection
- Human pose/gesture
- Tracking/estimation
- Scene classification
- Action recognition
- …



If you were tasked to build a model …

2. What do you need to **understand the joke**?

- Physical reasoning
- Social/emotional cues
- Intent inference/pragmatics
- Common-sense/world knowledge
- …



If you were tasked to build a model …

3. What do we need to understand the purpose of the image?

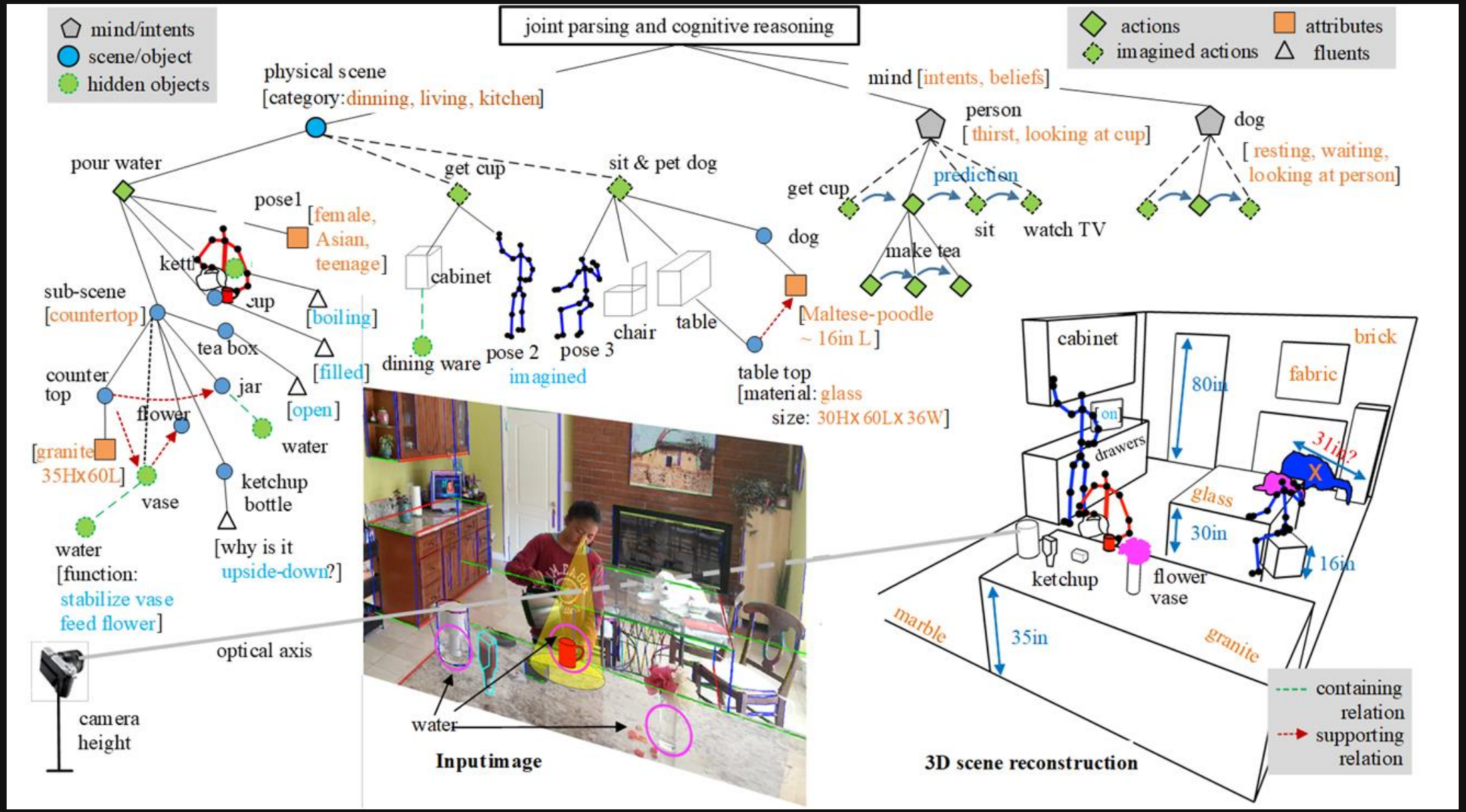
- Causal / physics understanding
- Theory-of-mind reasoning
- Social norm/pragmatics
- Humor schema
- …



Reasoning in typical computer vision tasks



Reasoning in typical computer vision tasks



Input



Object Detection
Scene Parsing

Understand
Relations

Output:
Robot Execution

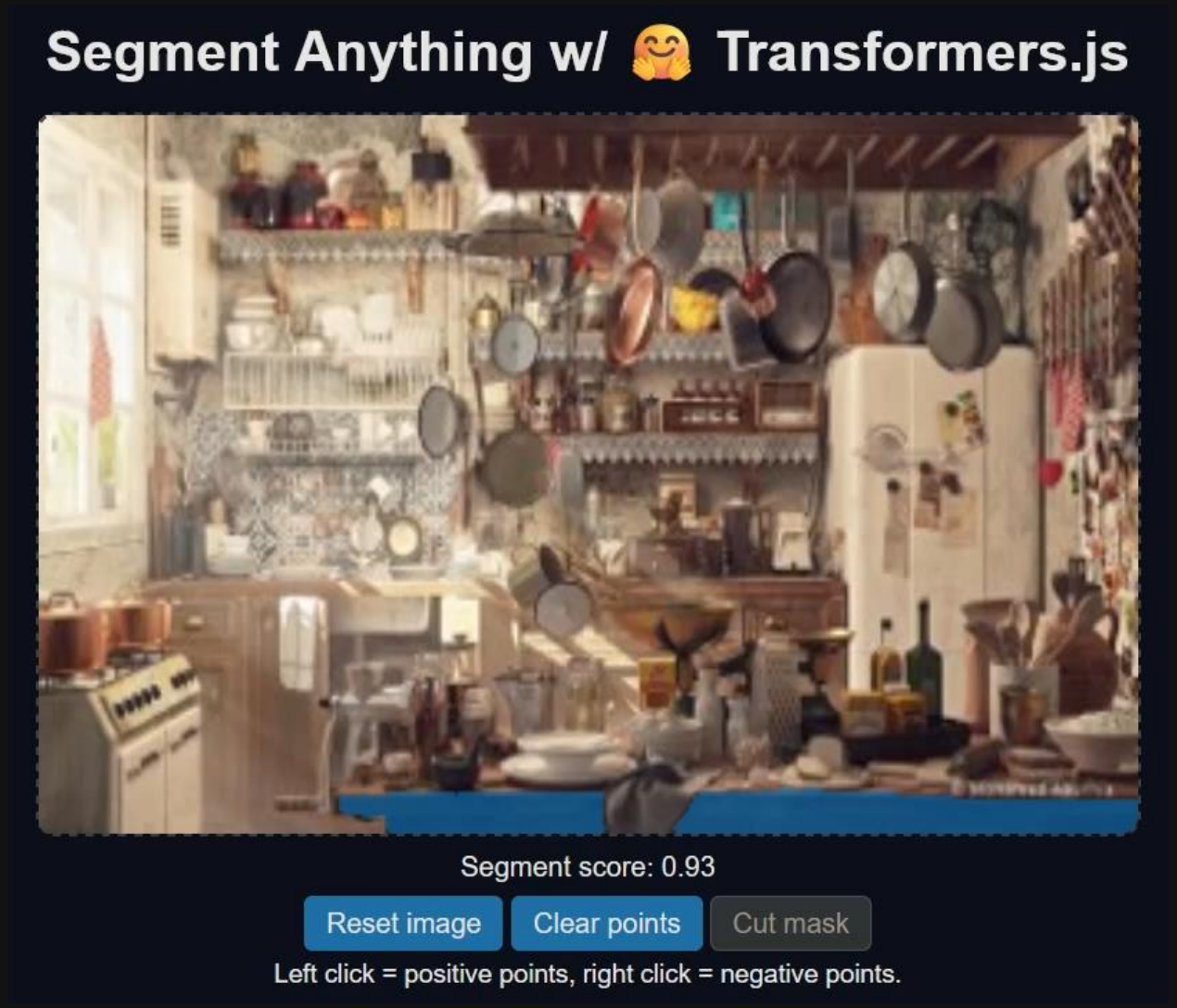
Pre-defined
Categorical Labels

HOIs

Humanoid


cookies
PEUS
© Mohamed AbuYhia

SOTA Segmentation



SOTA Object Detection (Small Model)



Input



Object Detection
Scene Parsing

Understand
Relations

Output:
Robot Execution

Pre-defined
Categorical Labels

HOIs

Humanoid

Challenge 1: Categorical labels do not associate directly with actions

Cup, Chair, Smartphone, Rock/stone, Door handle, Notebook (paper), Spoon, Rubber band, Cardboard box

Input



Object Detection
Scene Parsing

Understand
Relations

Output:
Robot Execution

Pre-defined
Categorical Labels

HOIs

Humanoid

Challenge 2: Long-tail problems

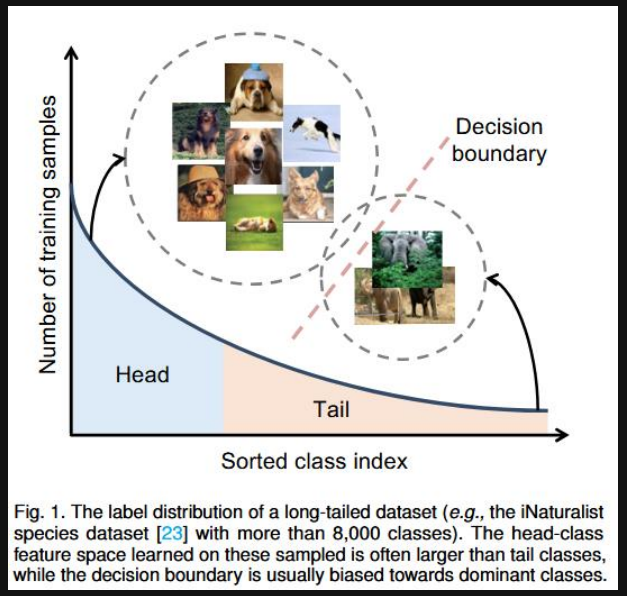


Fig. 1. The label distribution of a long-tailed dataset (*e.g.*, the iNaturalist species dataset [23] with more than 8,000 classes). The head-class feature space learned on these sampled is often larger than tail classes, while the decision boundary is usually biased towards dominant classes.

Input



Object Detection
Scene Parsing

Understand
Relations

Output:
Robot Execution

Possibility for
Actions

HOIs

Humanoid

Input



Object Detection
Scene Parsing

Understand
Relations

Output:
Robot Execution

Possibility for Actions

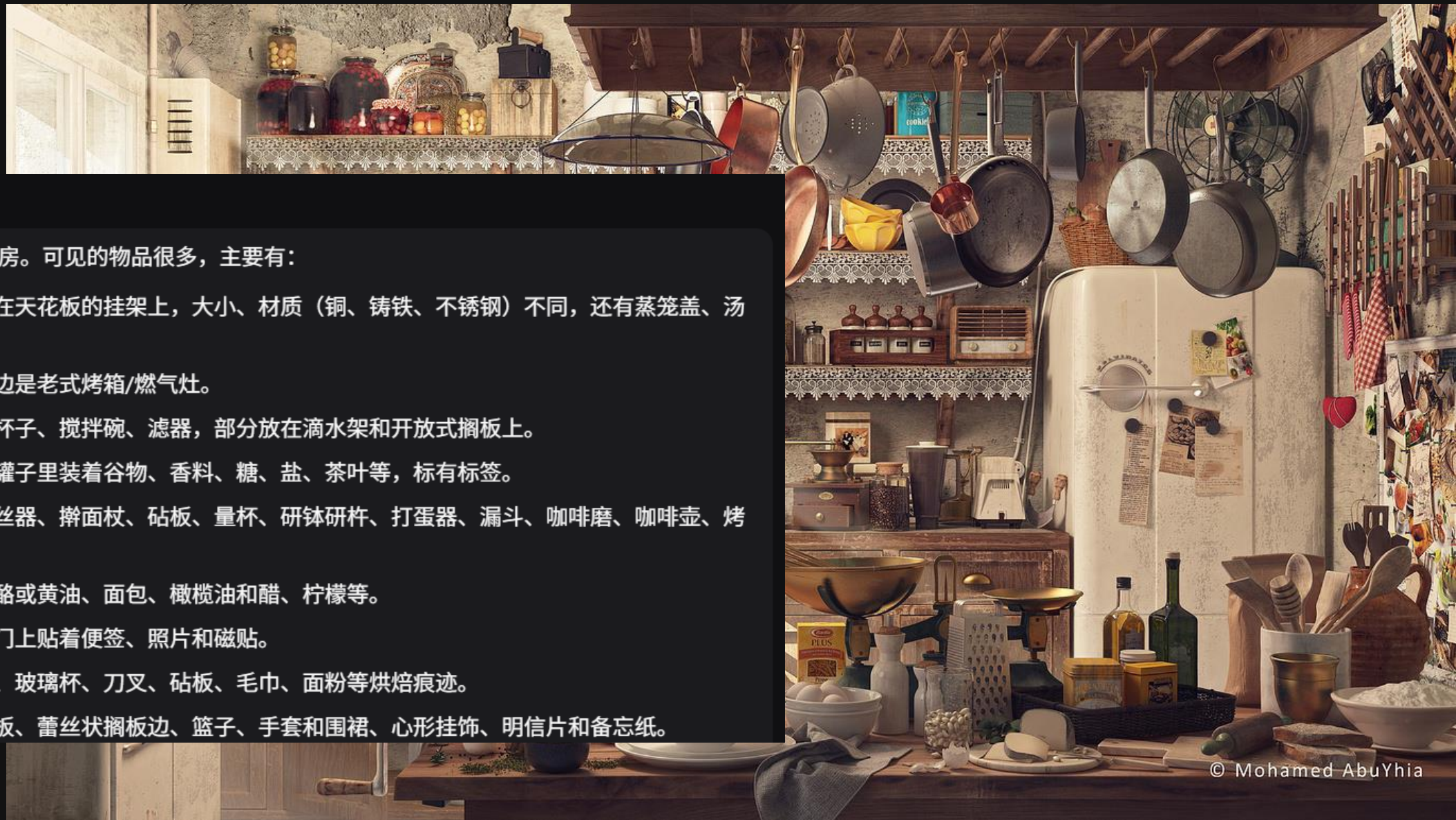
HOIs

Humanoid

- Cup: affords grasping, containing liquids, drinking, hanging by the handle, measuring roughly, making sound when tapped.
- Chair: affords sitting, stepping to reach high places, leaning against, blocking a door, hanging a bag on the backrest.
- Smartphone: affords calling, photographing, lighting (flashlight), navigation, paying, scanning codes, audio recording.
- Rock/stone: affords gripping, throwing, paperweight use, stepping on, grinding/scraping, striking to spark (with proper materials).
- Door handle (bar type): affords pushing/pulling, hanging items, using as a leverage point, signaling "this side opens."
- Notebook (paper): affords writing, sketching, folding, bookmarking, tearing into notes, fanning.
- Spoon: affords scooping, stirring, measuring small amounts, serving, reflecting as a makeshift mirror.
- Rubber band: affords bundling, fastening, gripping aid on a jar lid, temporary sling/ergonomic loop.
- Cardboard box: affords storing, carrying, stacking, sitting (short term), organizing, play/pretend structures.

Categorical Labels to Action Labels are N-to-N!!

SOTA Object Detection (VLM)



GPT-5

这是一间杂而温馨的复古厨房。可见的物品很多，主要有：

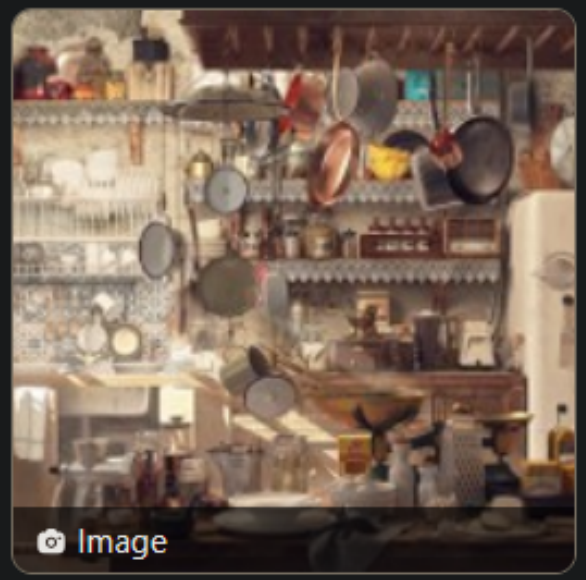
- 各种锅具和平底锅：吊在天花板的挂架上，大小、材质（铜、铸铁、不锈钢）不同，还有蒸笼盖、汤勺、筛网等。
- 炉灶上有铜色汤锅；旁边是老式烤箱/燃气灶。
- 大量餐具：碗、盘子、杯子、搅拌碗、滤器，部分放在滴水架和开放式搁板上。
- 调料和干货：玻璃瓶、罐子里装着谷物、香料、糖、盐、茶叶等，标有标签。
- 厨房工具：绞肉机、刨丝器、擀面杖、砧板、量杯、研钵研杵、打蛋器、漏斗、咖啡磨、咖啡壶、烤面包机、秤等。
- 食材：鸡蛋、香草、奶酪或黄油、面包、橄榄油和醋、柠檬等。
- 冰箱：老式白色冰箱，门上贴着便签、照片和磁贴。
- 台面与餐桌：摆着碗盘、玻璃杯、刀叉、砧板、毛巾、面粉等烘焙痕迹。
- 墙面与装饰：花砖挡水板、蕾丝状搁板边、篮子、手套和围裙、心形挂饰、明信片和备忘纸。

SOTA Object Detection (VLM)

onnx/prepare_inputs_embeds_q4.onnx
20.50 MB / 990.68 MB 2.1%

onnx/language_model_q4f16.onnx
32.07 MB / 665.89 MB 4.8%

Results



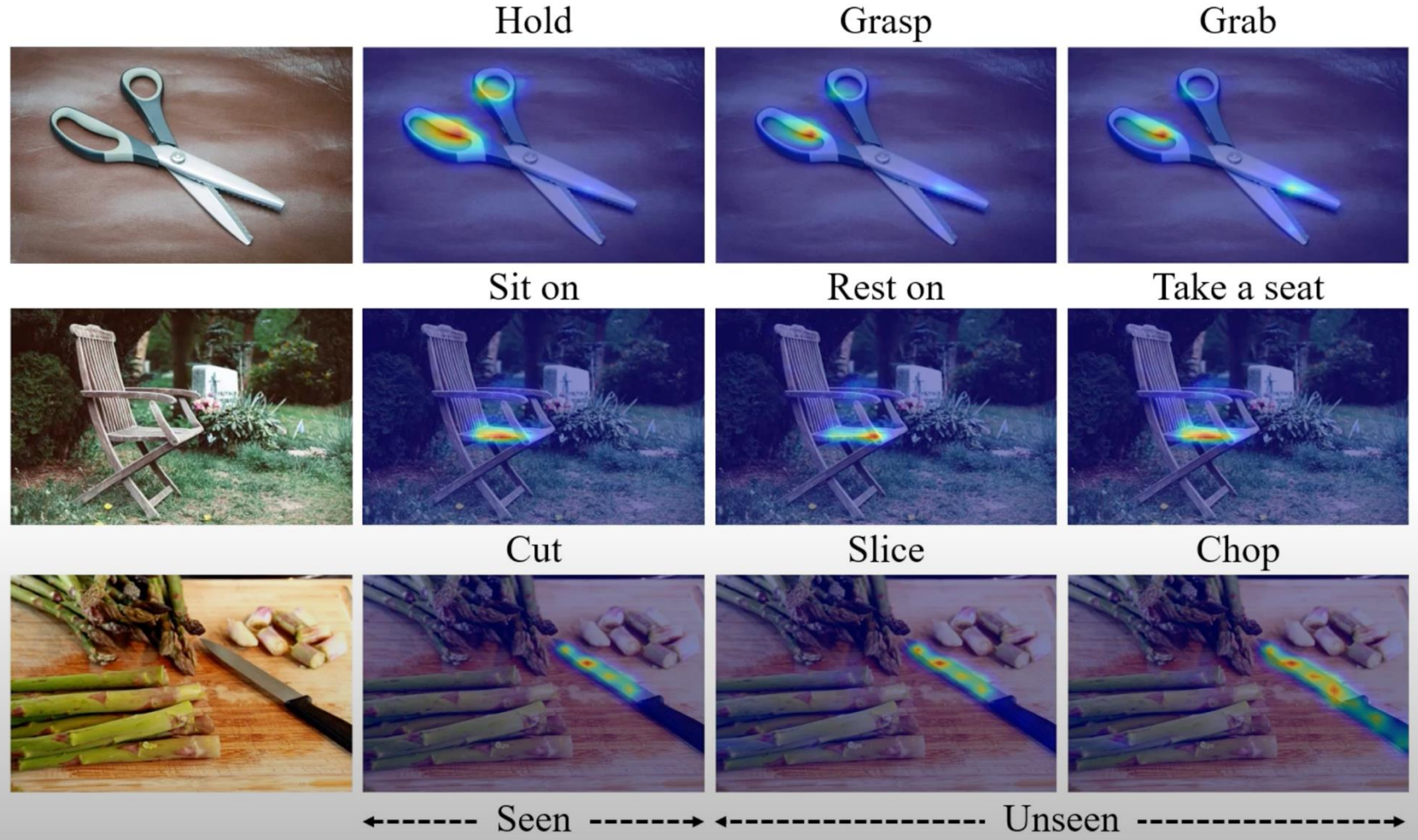
Complete

The image depicts a cluttered kitchen filled with various cooking utensils and kitchenware. The shelves are stocked with pots, pans, and dishes, while the countertops are cluttered with bowls, bottles, and other kitchen essentials. The kitchen appears to be well-used, with items such as a kettle, a toaster, and a bottle of olive oil scattered across the surfaces. The overall atmosphere suggests a busy and active cooking space, with a vintage refrigerator adorned with notes and magnets.

© Mohamed AbuYhia

Novel Affordance Results (AGD20K)

Li et al. One-shot open affordance learning with foundation models



Input



Object Detection
Scene Parsing

Understand
Relations

Output:
Robot Execution

Possibility for
Actions

HOIs

Humanoid

- Cup: affords grasping, containing liquids, drinking, hanging by the handle, measuring roughly, making sound when tapped.
- Chair: affords sitting, stepping to reach high places, leaning against, blocking a door, hanging a bag on the backrest.
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- Notebook (paper): affords writing, sketching, folding, bookmarking, tearing into notes, fanning.
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- Rubber band: affords bundling, fastening, gripping aid on a jar lid, temporary sling/ergonomic loop.
- Cardboard box: affords storing, carrying, stacking, sitting (short term), organizing, play/pretend structures.


ABB
FOUNDRY2
ABB

Input



Object Detection
Scene Parsing

Understand
Relations

Output:
Robot Execution

Possibility for
Actions
(Chapter 1)

HOIs
(Chapter 2)

Humanoid
(Chapter 3)

- Cup: affords grasping, containing liquids, drinking, hanging by the handle, measuring roughly, making sound when tapped.
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Chapter 1
Affordance

The opportunity for task and action



Human-made scenes afford various human activities


House of Yahweh







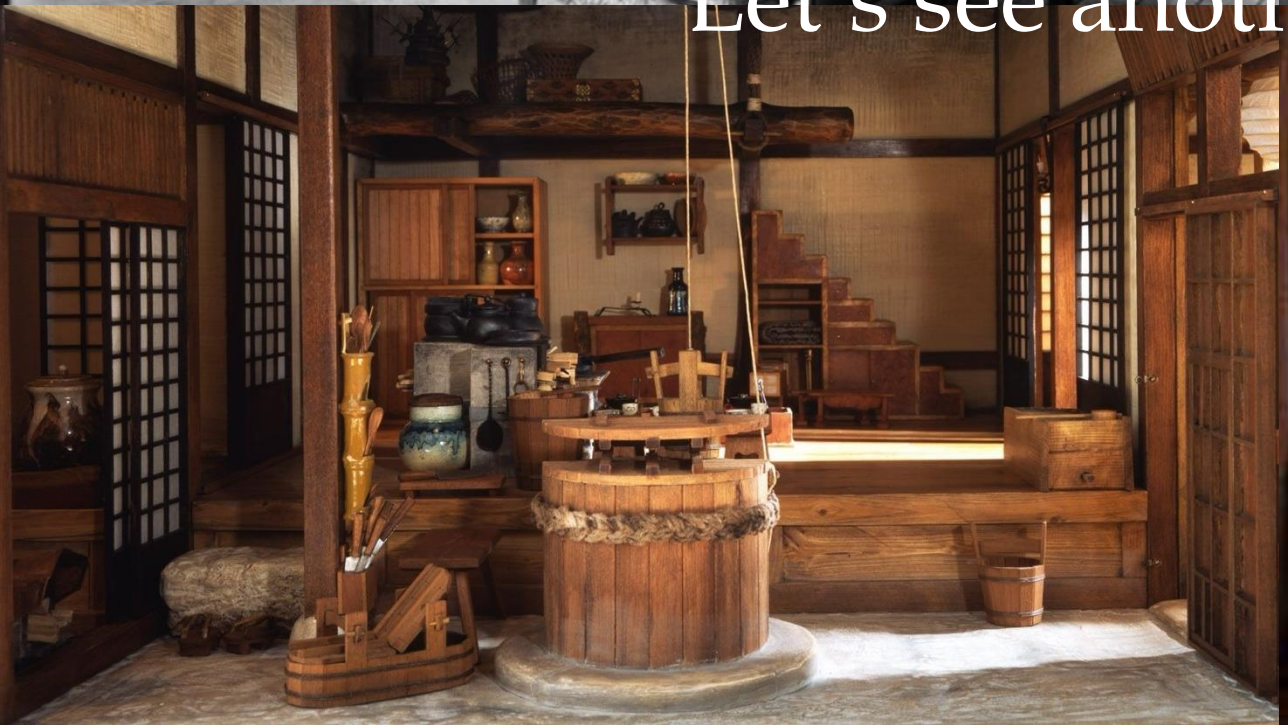
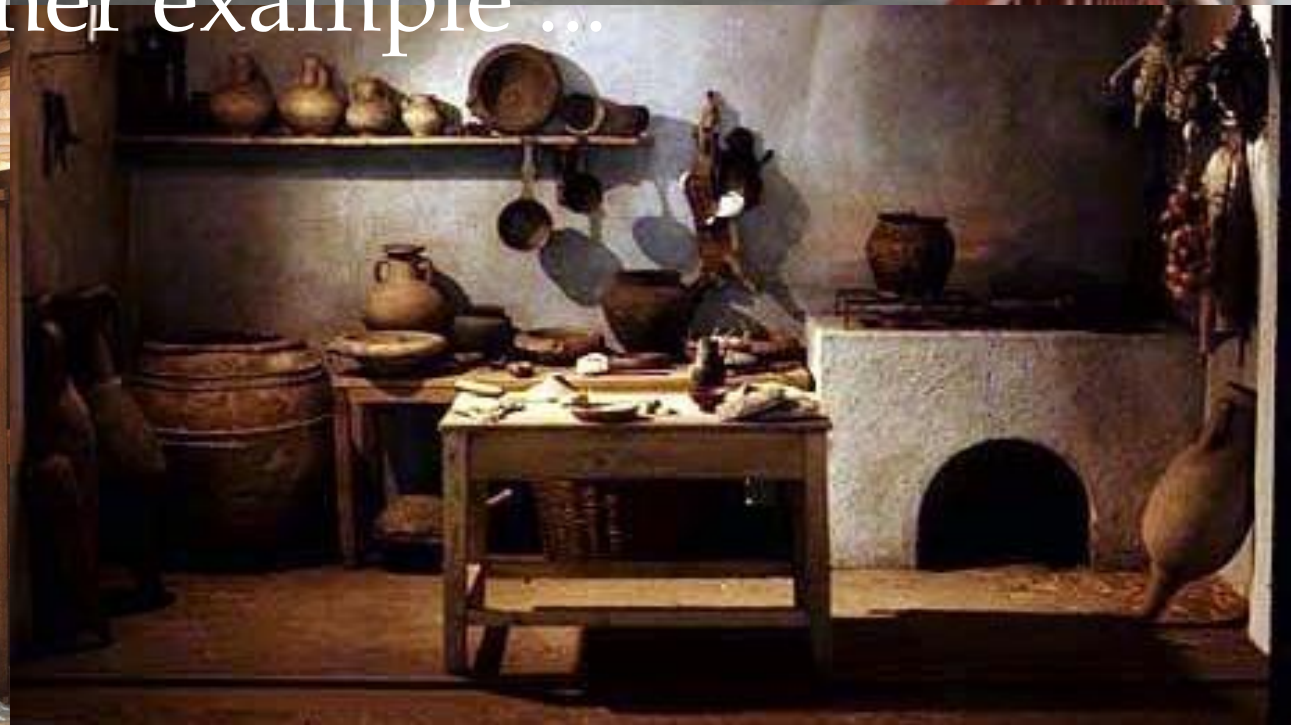


What makes you think these two rooms are kitchens?







Let's see another example ...

Affordance provides strong priors for us to effortlessly understand the most complex indoor environments





A simpler, one-object case




2025-09-02 23:22:50
一张床睡不出两种人

Which chair seems more comfortable?



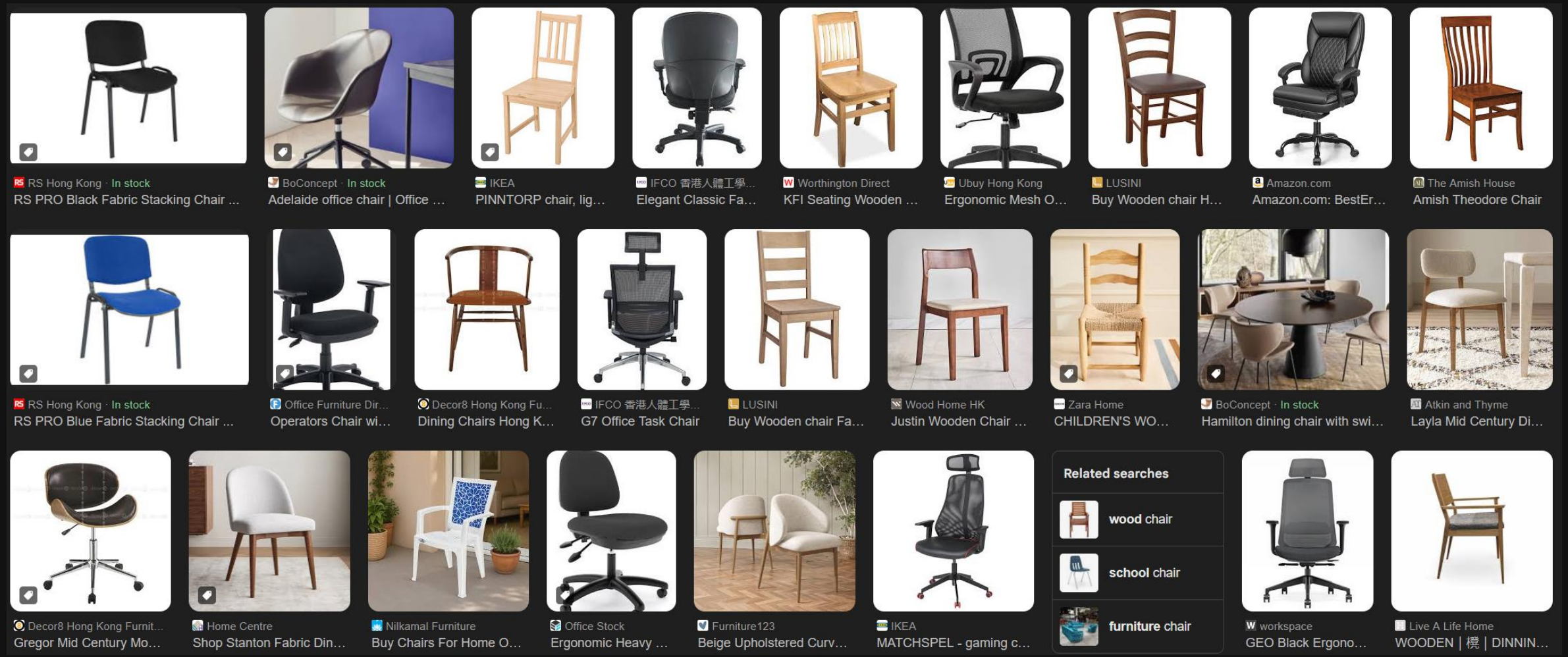


Different appearance/ geometry → Same category





Different appearance/ geometry → Same category



Outline

1. Cognitive bases for affordance
2. Computational bases for affordance
3. From affordance to functionality

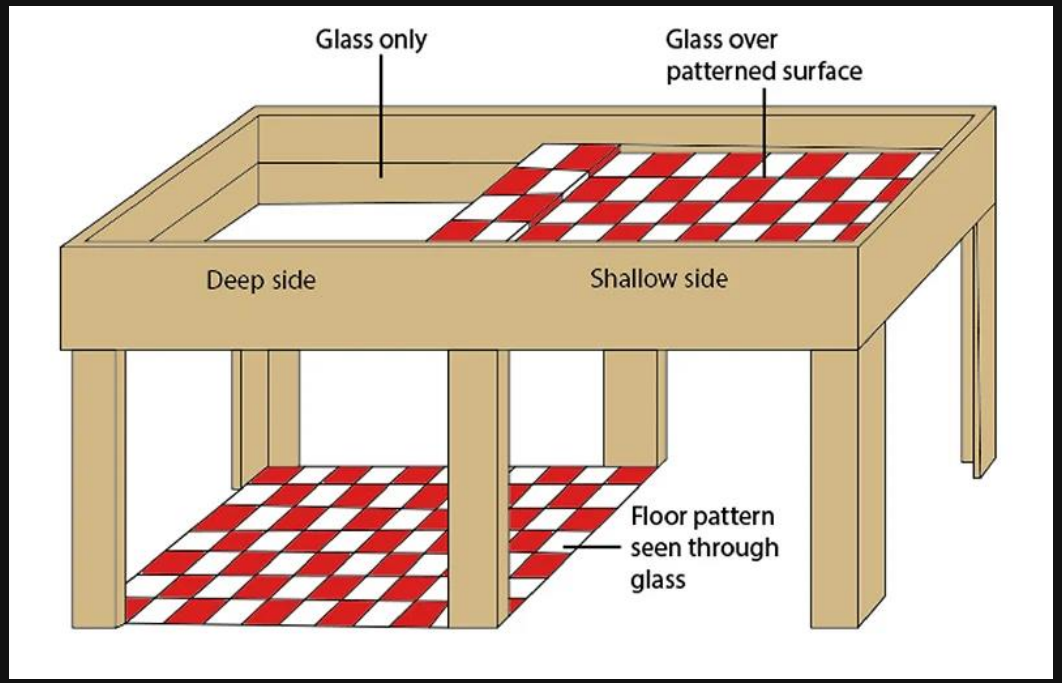
The Visual Cliff Experiment



The Visual Cliff Experiment

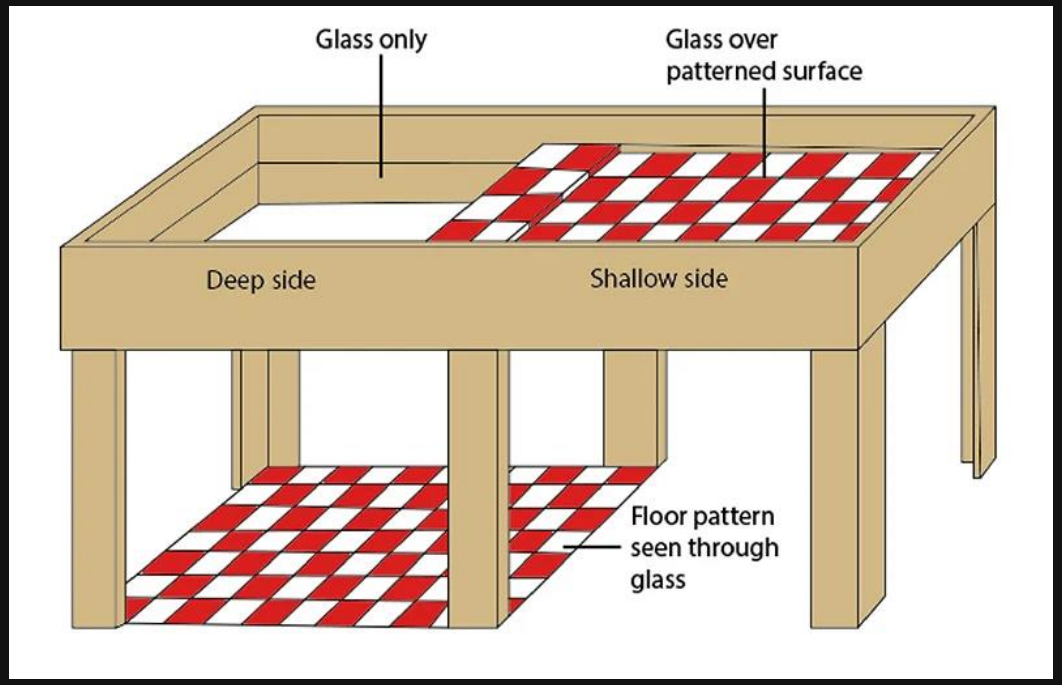
- A glass table with a checkered pattern underneath creates two sides - one "shallow" where the pattern sits directly under the glass, and one "deep" where the pattern is placed far below to simulate a cliff drop-off.

- Crawling infants (6-14 months old) are placed on the shallow side while their mothers call to them from across the apparent cliff on the other side.



The Visual Cliff Experiment

- Most infants refused to crawl over the visual cliff despite their mothers' encouragement, suggesting they could perceive the depth and recognized it as dangerous.



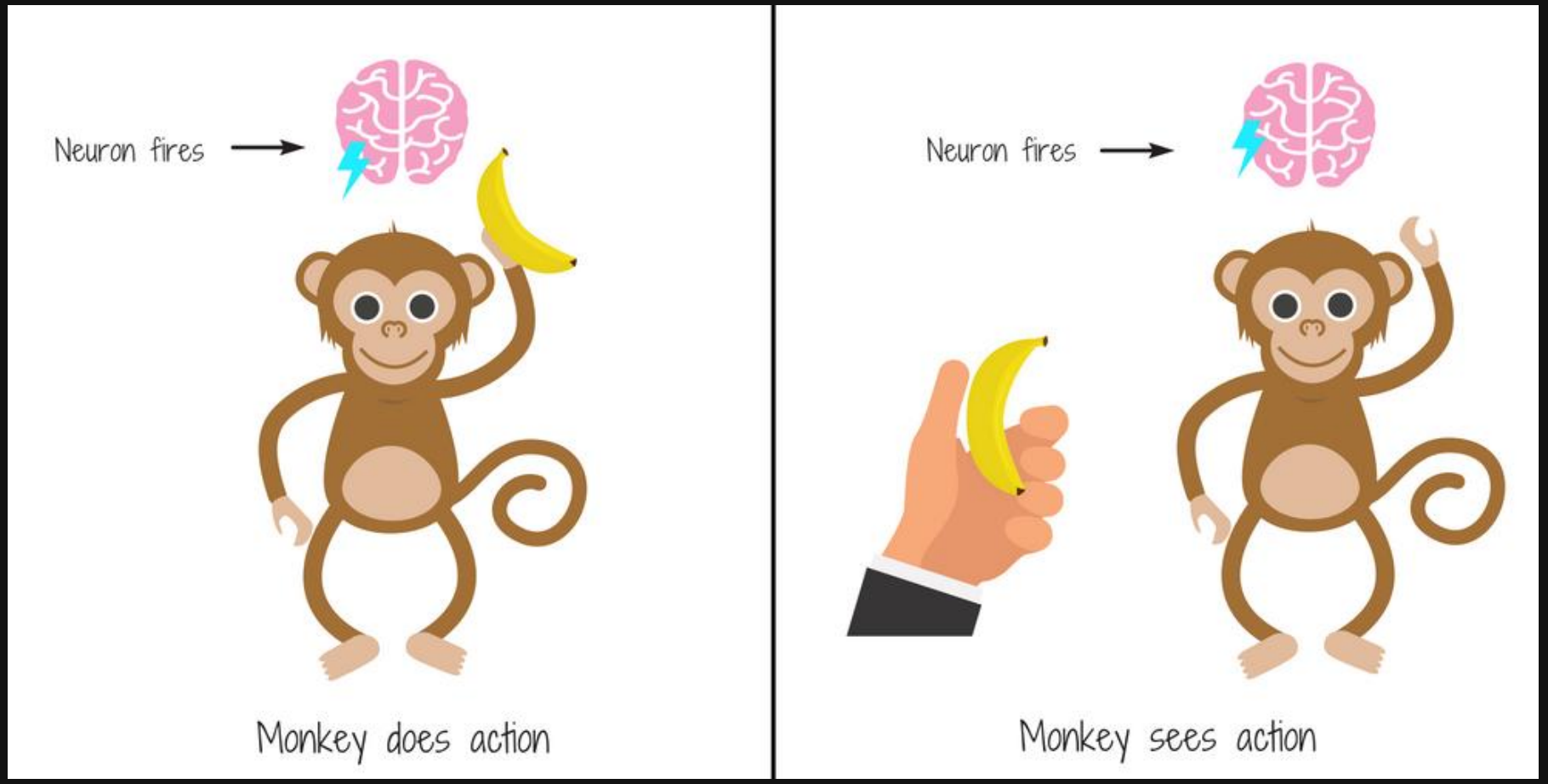
Infants aren't just perceiving depth as an abstract visual feature - they're perceiving what that depth means for their possible actions.

The Mirror Neuron Systems



The Mirror Neuron Systems

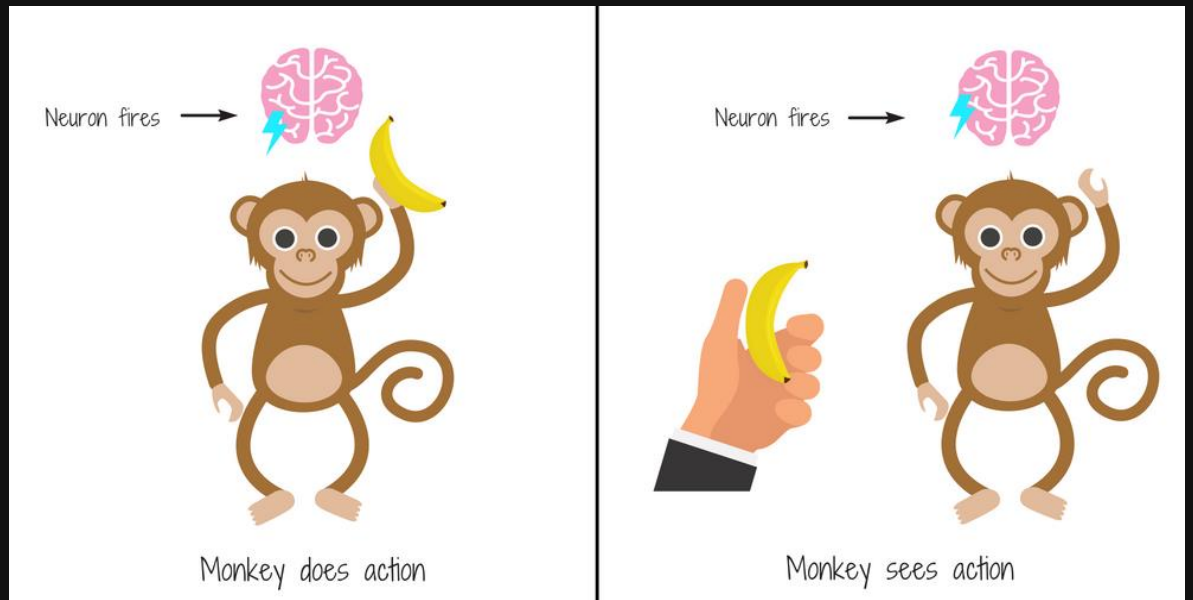
Mirror neurons fire both when an individual performs an action and when they observe another individual performing the same action.

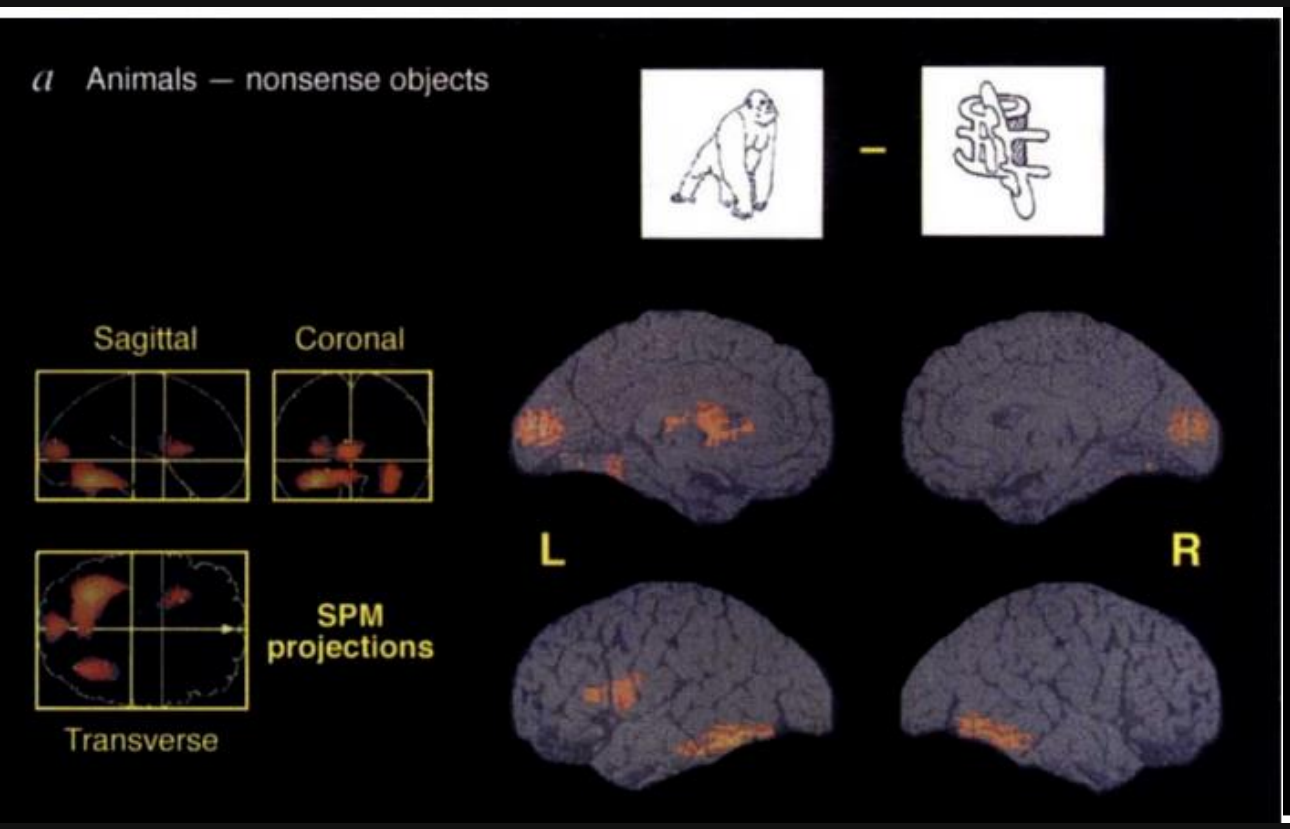


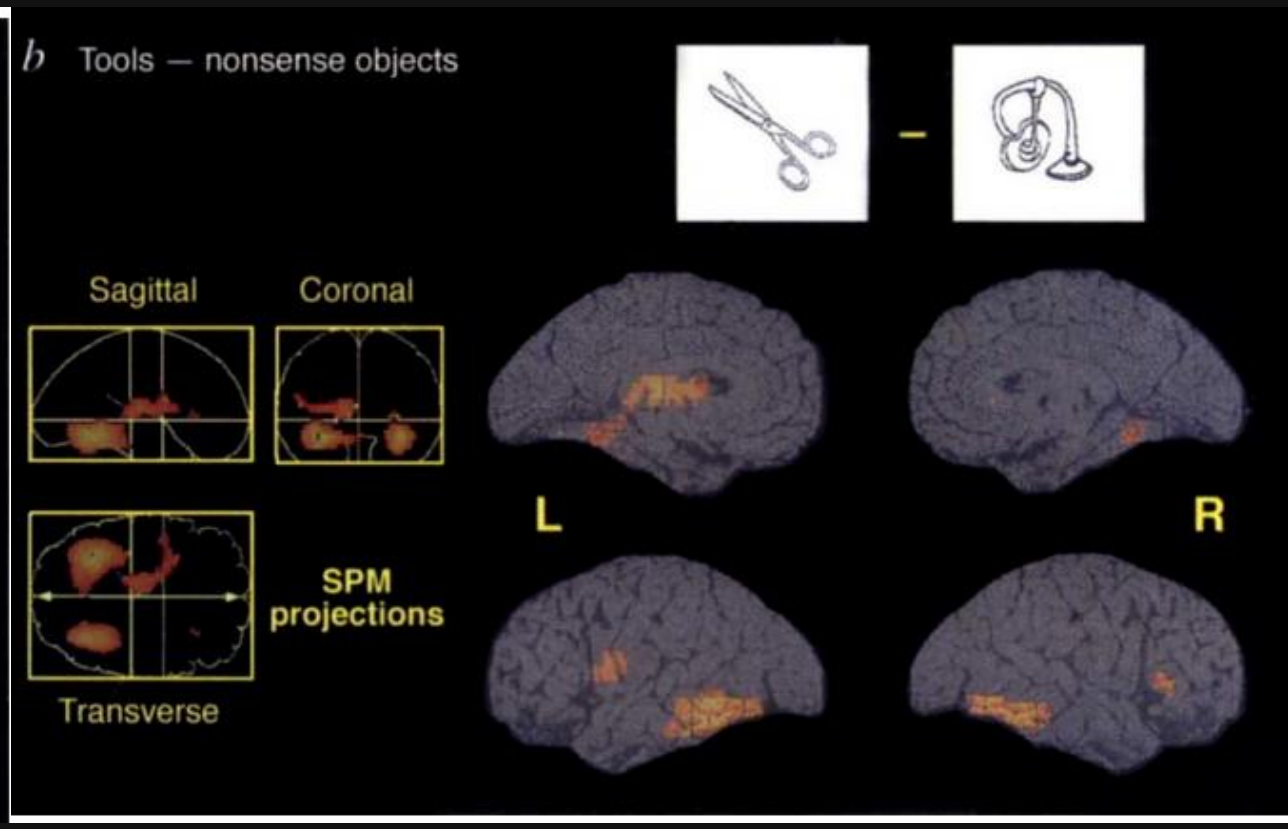
The Mirror Neuron Systems

Mirror neuron suggests that our perception of objects is inherently tied to potential actions we could perform with them.

When we see a hammer, mirror neurons associated with grasping and swinging actions may activate, essentially "simulating" the afforded actions even when we're just observing.





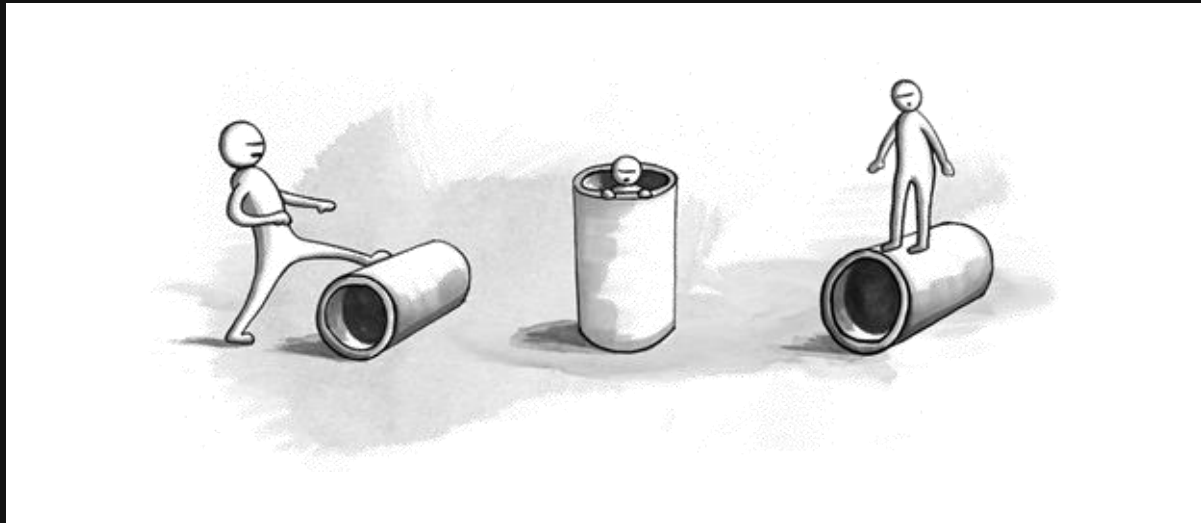


Martin, A., Wiggs, C. L., Ungerleider, L. G., & Haxby, J. V. (1996). Neural correlates of category-specific knowledge. Nature.

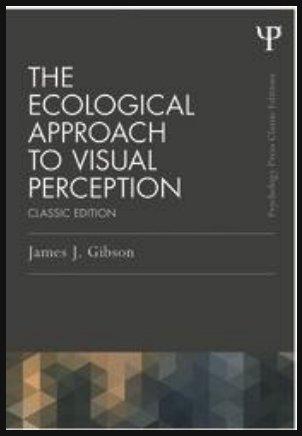
Definition of Affordance

The affordances of the environment are what it offers the animal, what it provides or furnishes, either for good or ill. The verb to afford is found in the dictionary, but the noun affordance is not. I have made it up. I mean by it something that refers to both the environment and the animal in a way that no existing term does. It implies the complementarity of the animal and the environment.

–James J. Gibson, 1979.







How can an object be used by an agent?



Gibsonian Affordances

Key Insight: Our perception defines the environment.

Gibsonian Affordances

- The Medium: Air affords breathing/sound transmission/ …

Gibsonian Affordances

- The Medium: Air affords breathing/sound transmission/…
- The Substances: Water affords drinking/pouring. Solids afford manufacture.

Gibsonian Affordances

- The Medium: Air affords breathing/sound transmission/…
- The Substances: Water affords drinking/pouring. Solids afford manufacture.
- The Surfaces and their Layouts: Horizontal, flat, extended, rigid surfaces afford support. Vertical, flat, extended, rigid surfaces afford barrier. Slopes afford walking or climbing.

Gibsonian Affordances

- The Medium: Air affords breathing/sound transmission/…
- The Substances: Water affords drinking/pouring. Solids afford manufacture.
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- The Objects: Extremely various. What we perceive when we look at objects are their affordances.

Gibsonian Affordances

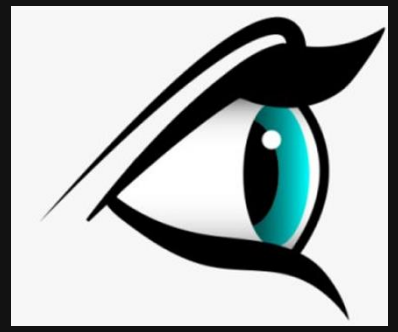
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- The Surfaces and their Layouts: Horizontal, flat, extended, rigid surfaces afford support. Vertical, flat, extended, rigid surfaces afford barrier. Slopes afford walking or climbing.
- The Objects: Extremely various. What we perceive when we look at objects are their affordances.
- Other Persons and Animals: When touched they touch back, when struck they strike back; in short, they interact with the observer and with one another.

Let's look at an example



Let's look at an example

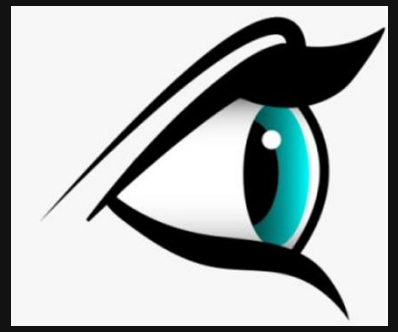






Let's look at an example



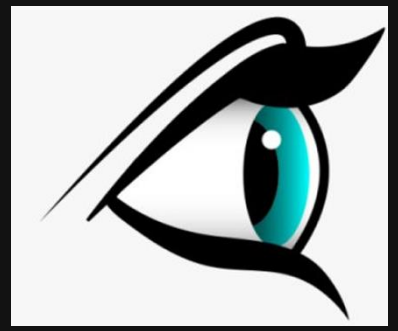




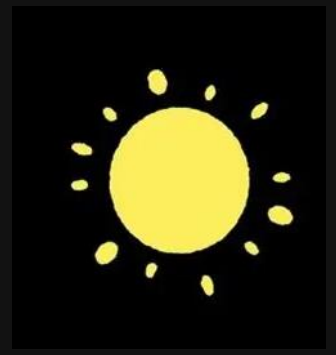
Let's look at an example



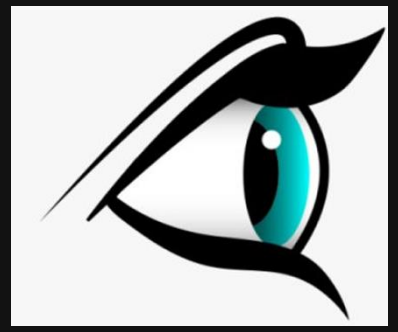




Let's look at an example



horizontal, flat,
wide, rigid



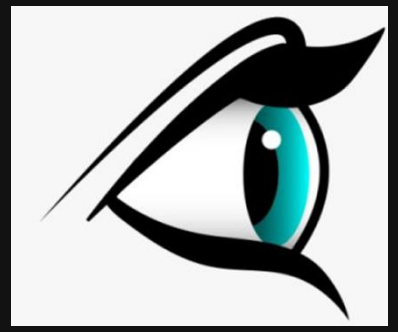


Let's look at an example



sittable

horizontal, flat, wide, rigid





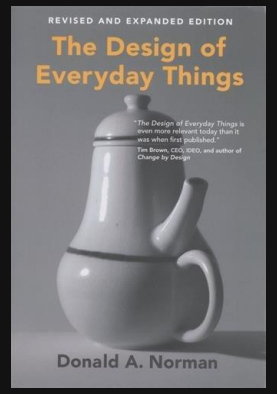
Let's look at an example



- Gibson argues:
  - If perceiving light tells us an object is:
    - flat
    - rigid
    - horizontal
    - wide
- Then perceiving light also tells us an object is sittable
- There is no separate, distinct, cognitive loop involved in parsing perceived physical properties and deciding whether or not something can be sat on

Don Norman, a cognitive scientist, explained affordance using everyday things in 1994

"The value of a well-designed object is when it has such a rich set of affordances that people who use it can do things with it that the designer never imagined."



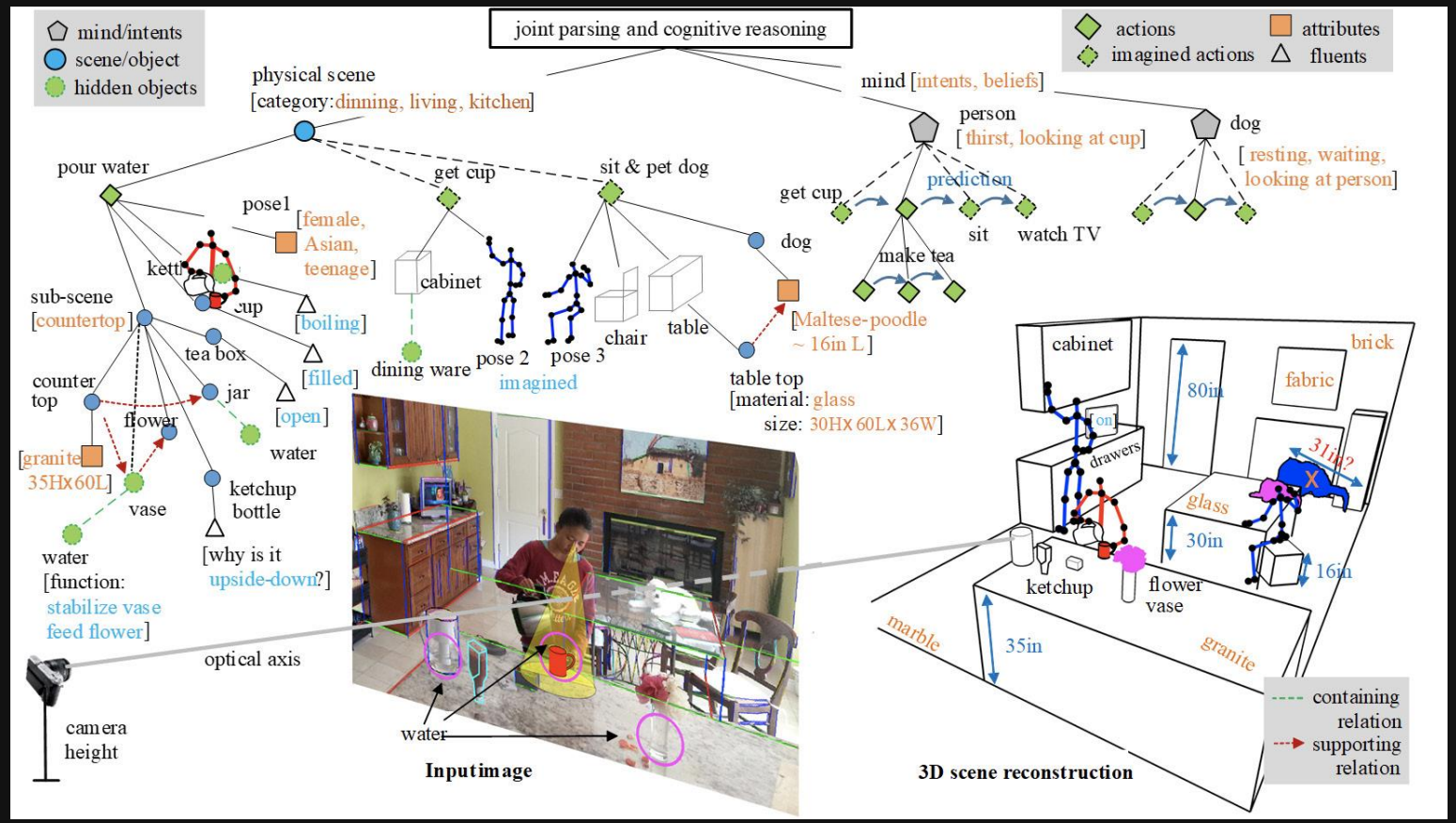




Outline

1. Cognitive bases for affordance
2. Computational bases for affordance
3. From affordance to functionality

Affordance: Beyond Object Detection



Gibson advocated that the target of computer vision should be to estimate possible interactions between a human, animal and environment in a scene rather than merely detecting the contents of a scene.

– Hassanin et al., 2021.

Perceive an affordance ≠ Classify an object

The theory of affordances rescues us from the philosophical muddle of assuming fixed classes of objects, each defined by its common feature and then give a name . . . You do not have to classify and label things in order to perceive what they afford . . . It is never necessary to distinguish all the features of an object, and in fact, it would be impossible to do so."

−James J. Gibson, 1979.

Yes, it is a stone.
But you will first
perceive it as a …



bookend



hammer

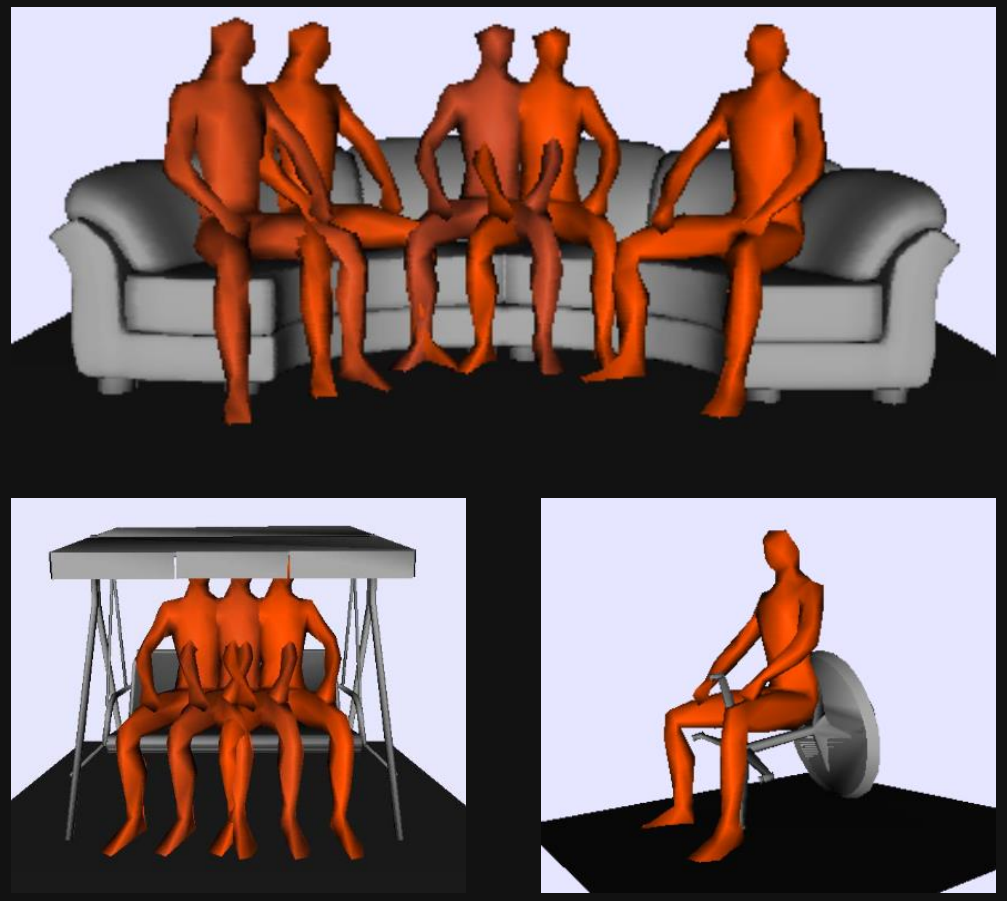


cairn

What makes a chair a chair…



…as long as it is sittable



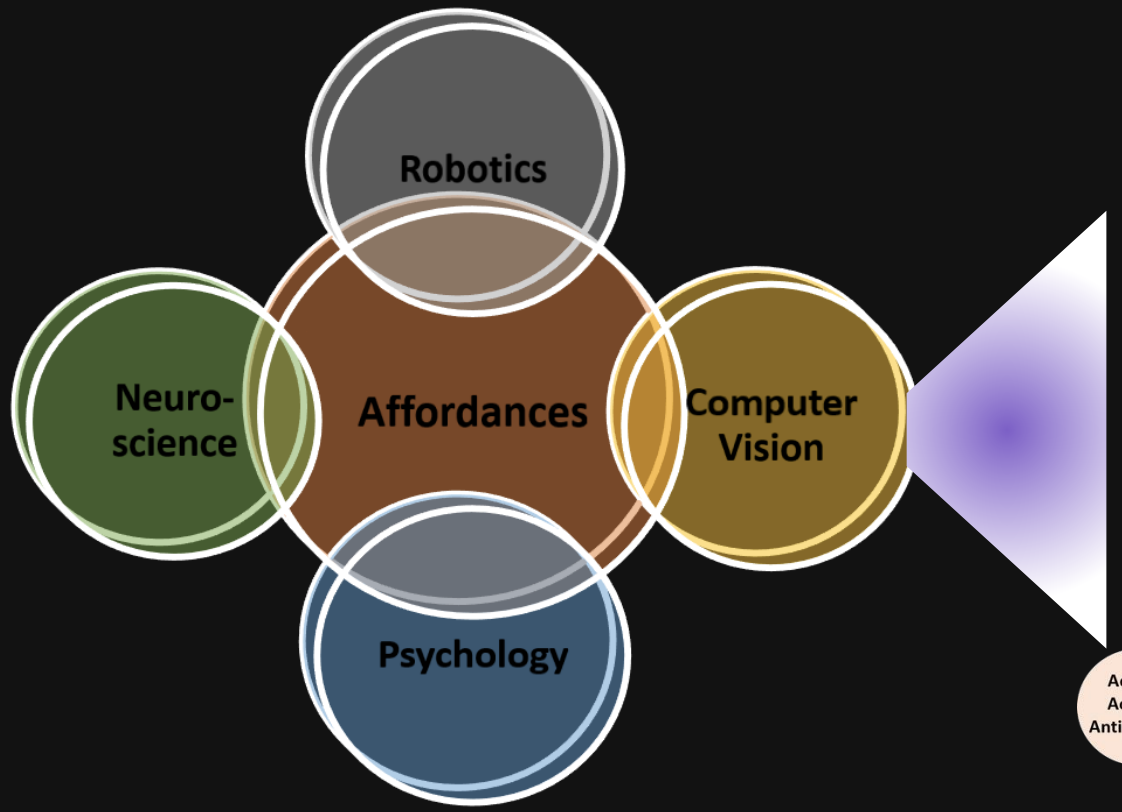
Grabner, Helmut, Juergen Gall, and Luc Van Gool. "What makes a chair a chair?." CVPR 2011. IEEE, 2011.

From Perception to Action: Computational Affordance Learning

A General Picture of Affordance Research

Hassanin et al., 2021.

Affordance Reseach is Cross-Disciplinary



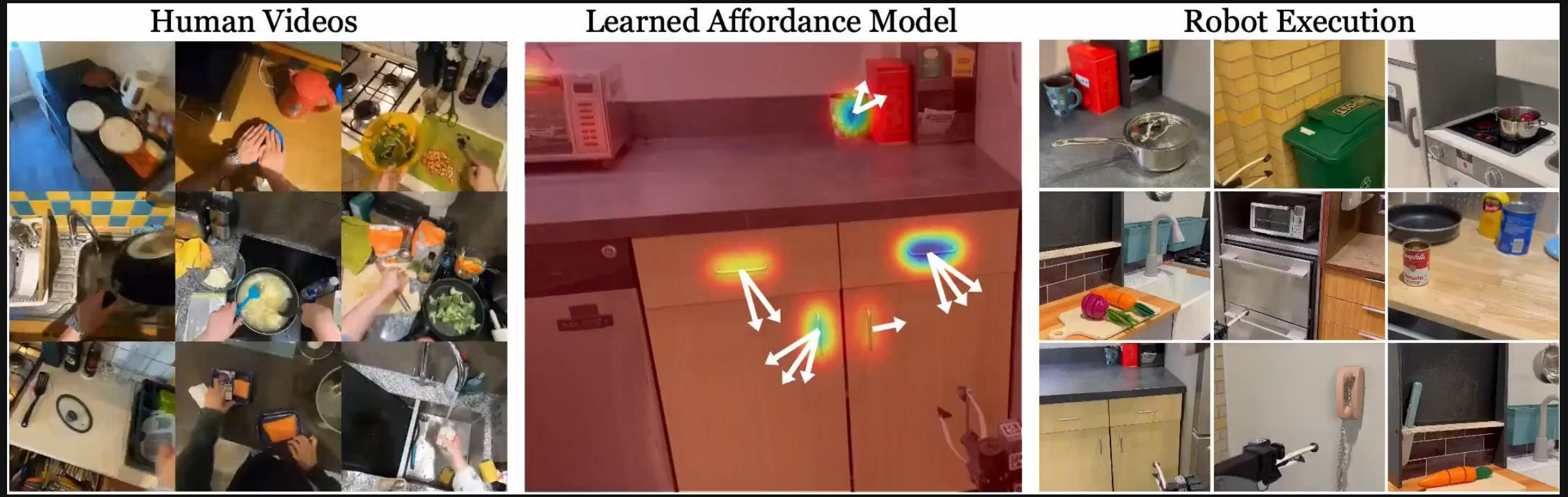
Affordances and Other Fields



Relation between Affordances and Visual Fields

Affordance Reseach is Cross-Disciplinary

Affordance can serve as mid-level representations from human demonstrations to robot execution



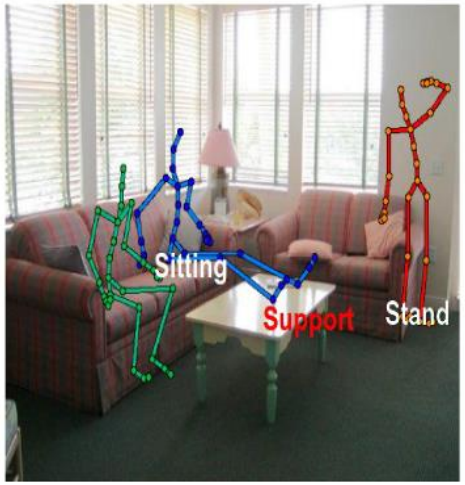
Bahl S, Mendonca R, Chen L, et al. Affordances from human videos as a versatile representation for robotics[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023: 13778-13790.

Affordances in Various Forms

From detection, segmentation, and categorization to reasoning



(a) Functionality understanding



(b) Contextual Affordances



(c) Affordance detection



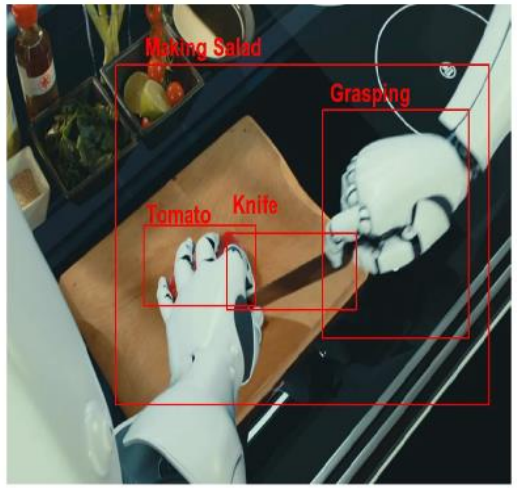
(d) Affordance categorization



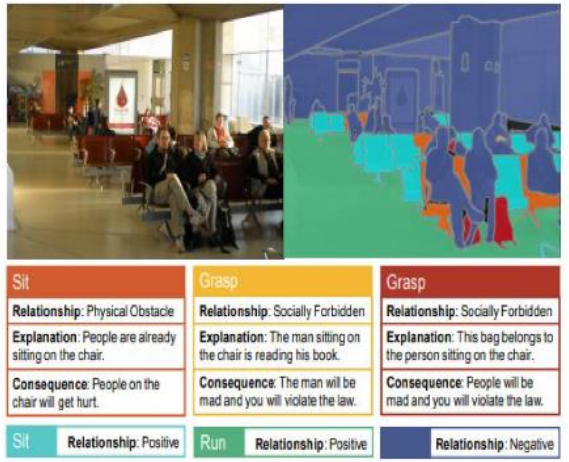
(e) Affordance segmentation



(f) Affordance reasoning



(g) Affordance-based activity



(h) Social affordances[14]



Multiple sources of features that can be fused together to create the affordance context.

All of which makes affordance learning a non-trivial problem

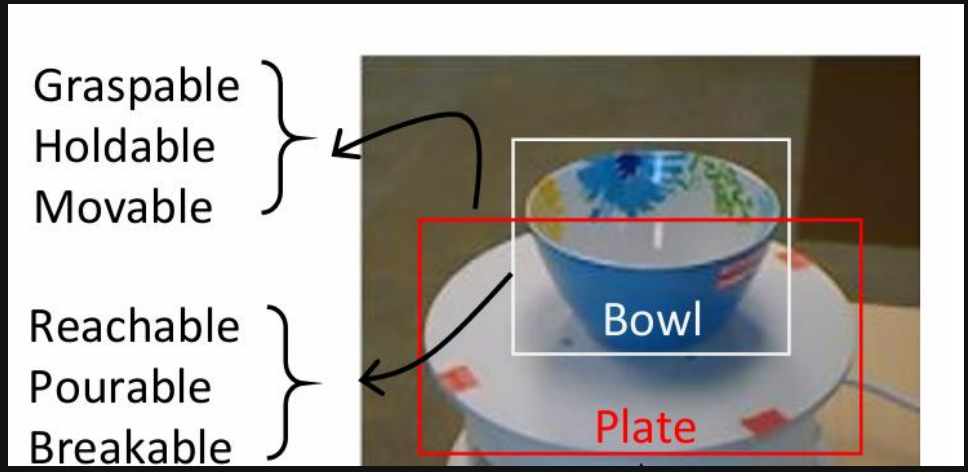
Typical Tasks in Affordance Learning

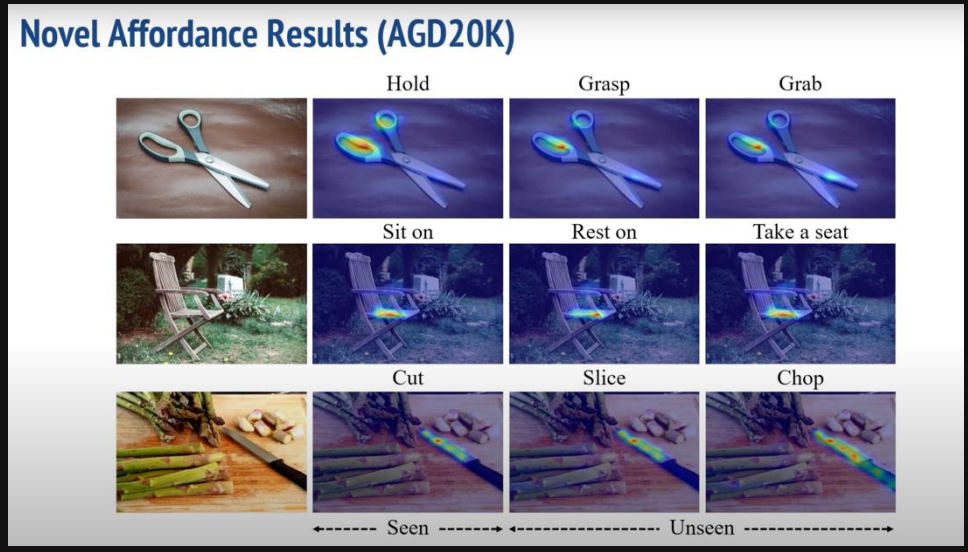
- Affordance Detection/Segmentation
- Scene Affordance
- Social Affordance
- …

Typical Tasks in Affordance Learning

- Affordance Detection/Segmentation
- Scene Affordance
- Social Affordance
- …

2D affordance detection / segmentation shares a similar formulation as image detection / segmentation





Affordance Detection

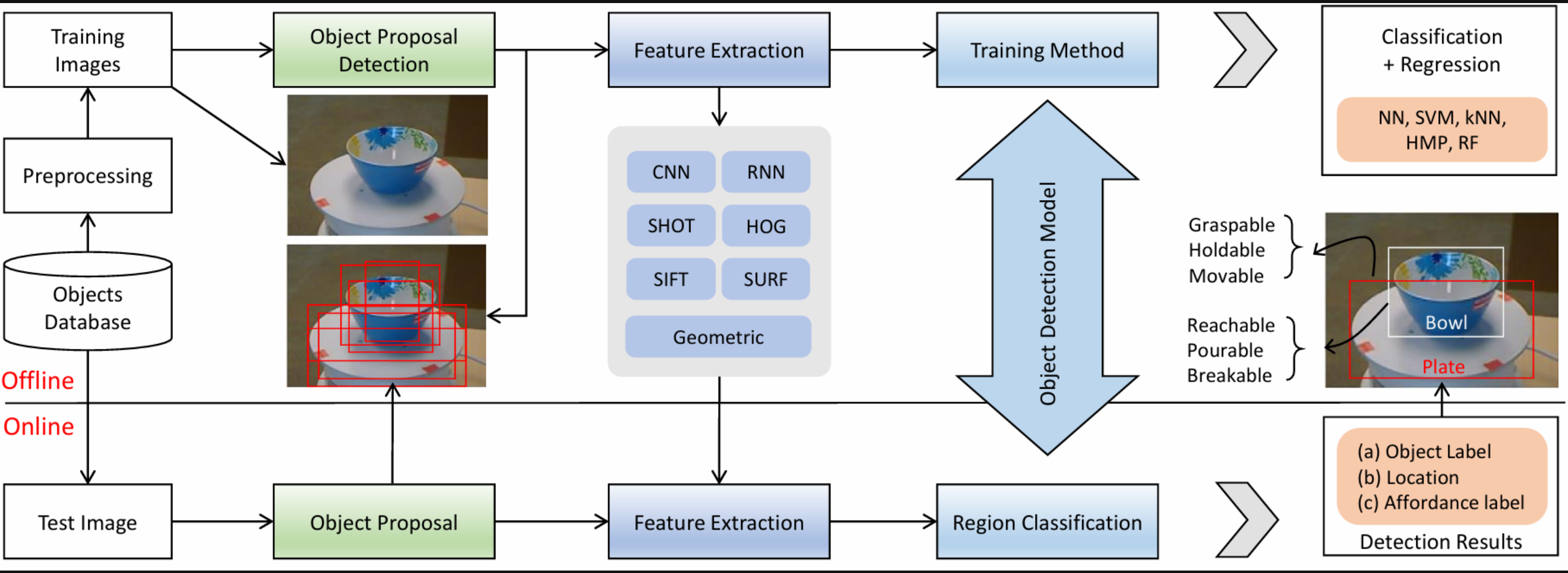
- Similar to object detection, affordance detection localizes and labels the affordance for scene objects. Different from conventional detection tasks, it targets only the salient objects that are most relevant for actions (instead of all objects in the scene).
- Find a mapping $f: X \rightarrow Y$
  - $X = \{x_1, x_2, \ldots, x_n\}$ denotes the set of candidate bounding boxes
  - $Y = \{y_1, y_2, \ldots, y_n\}$ denotes the affordance label of each bounding box

Affordance Segmentation

- Affordance segmentation segments an image into a set of regions which are labeled with a semantically meaningful affordance category. Remarkably, this task assigns a category label to each pixel in a region of interest.
- Find a mapping $f: P \rightarrow Y$
  - $P = \{p_1, p_2, \ldots, p_n\}$ denotes the set of n image pixels
  - $Y = \{y_1, y_2, \ldots, y_n\}$ denotes the affordance label of each image pixel

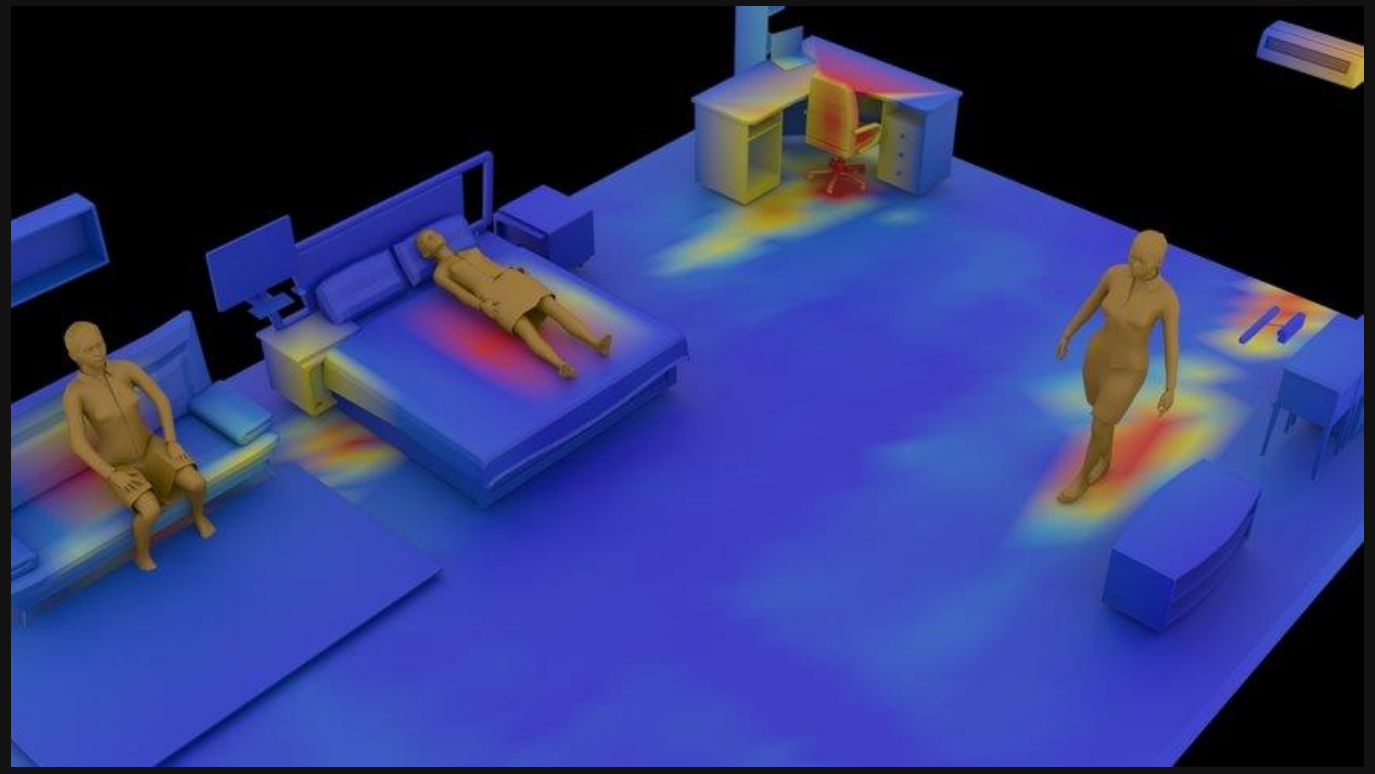
Method

With traditional computer vision paradigm, we can extract proper features and learn affordance distribution



Typical Tasks in Affordance Learning

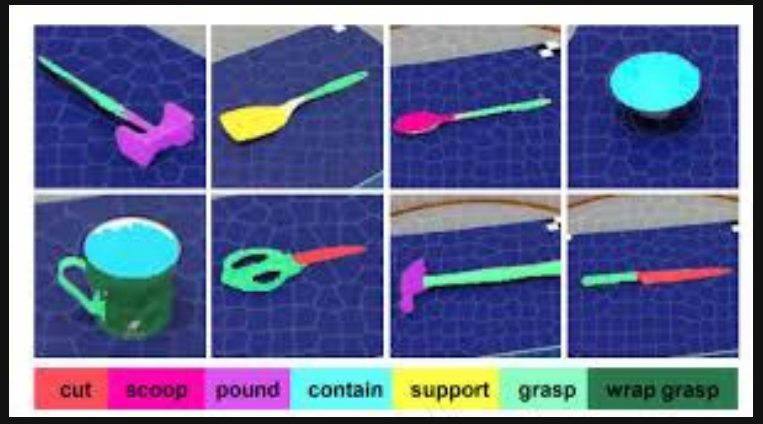
- Affordance Detection/Segmentation
- Scene Affordance
- Social Affordance
- …



Scene Affordance

- Scene affordance learning aims to predict what actions are possible across an entire scene rather than for individual objects, considering spatial relationships, multi-object interactions, and environmental constraints that emerge at the scene level.

Object Affordance vs Scene Affordance





- Object Affordance
  - potential interactions of objects.
  - based on objects properties.
    - shape, size, articulation, etc.

Scene Affordance

- potential interactions within a broader context or environment.
- based on combination and arrangement of multiple objects.

Case Study: Put Humans in the Scene

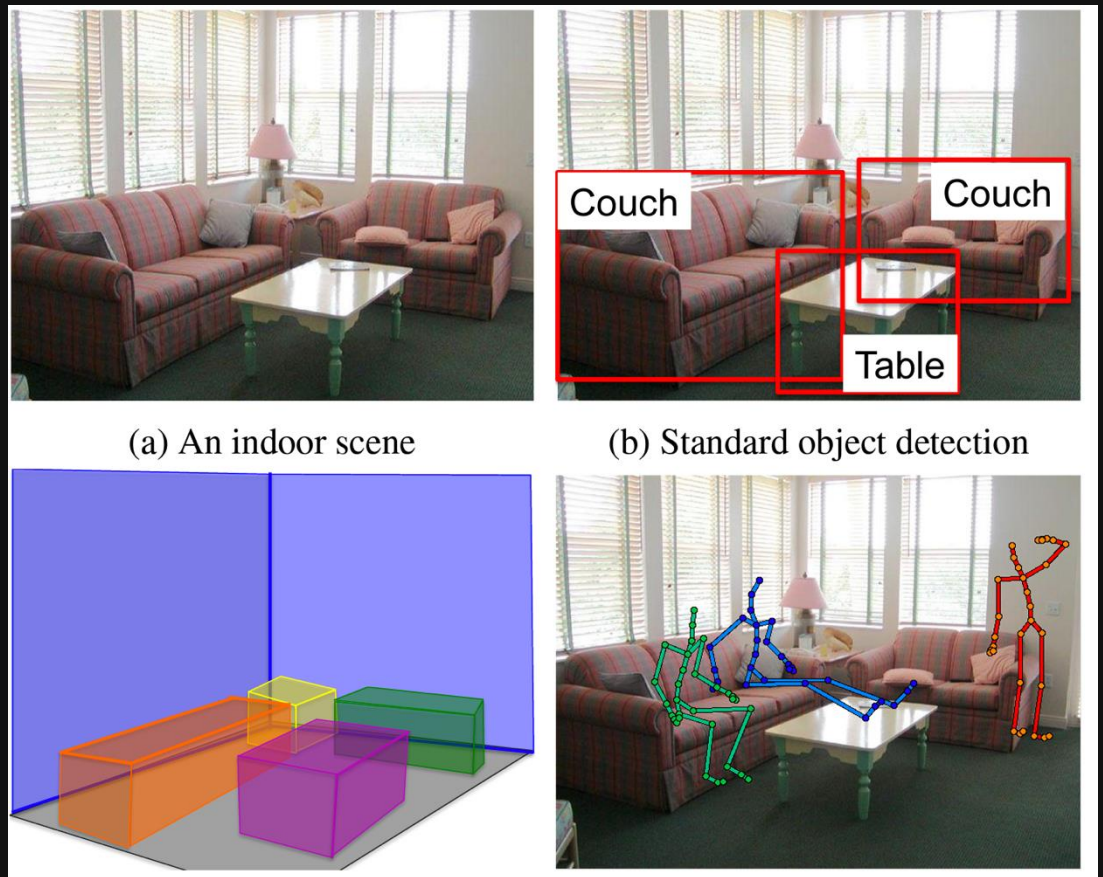
- Learning scene affordance by understanding how to “place” a human in the scene.
- “Human-made scenes are functional spaces that serve human activities, whose objects exist primarily to assist human actions.”

— — J. J. Gibson

A Short Demo



Task: Predict the full human workspace of the scene from a single image



(a) An indoor scene (b) Standard object detection

(c) Geometry estimation (d) Our human-centric representation

Figure 1: There are many ways to "understand" an image; *e.g.,* by naming the objects within it (b), or by estimating the scene geometry (c). We present a human-centric scene representation that models the human workspace of a scene. It can predict human poses afforded by the geometry in a given image (d).

Approach

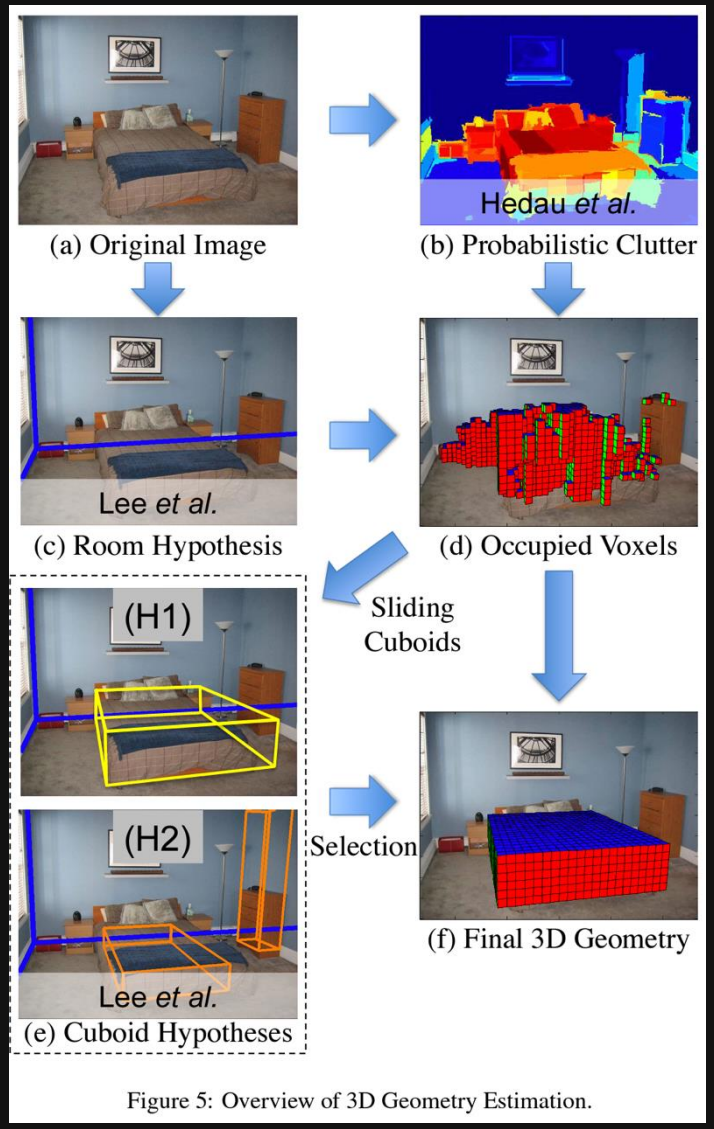


Figure 5: Overview of 3D Geometry Estimation.

Hard-coded Poses!

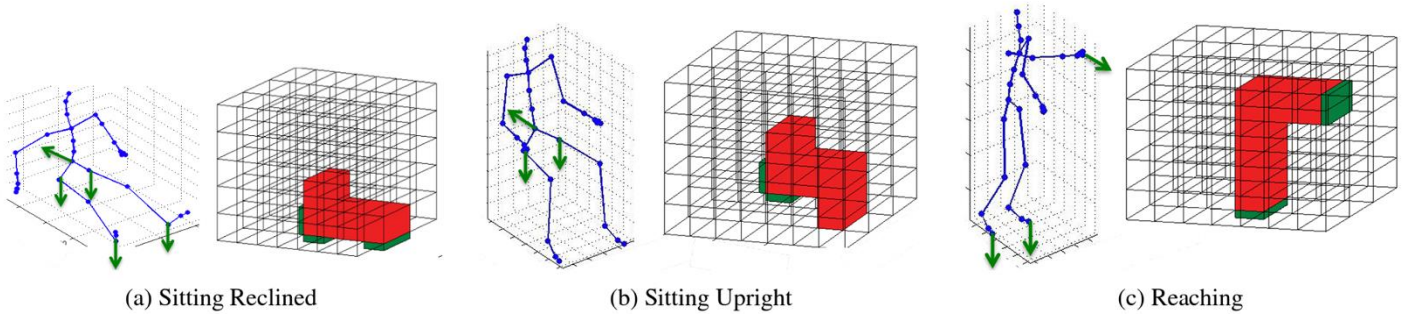


Figure 2: Qualitative Representation of Human Poses: Each pose is represented by the occupied blocks in discretized 3D space (shown in red) and the required surfaces of interaction (shown in green).

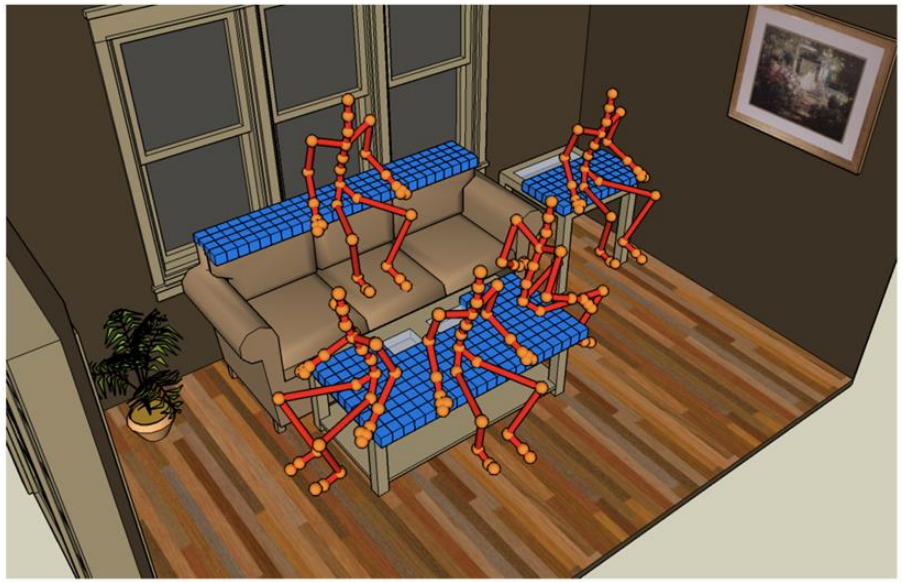
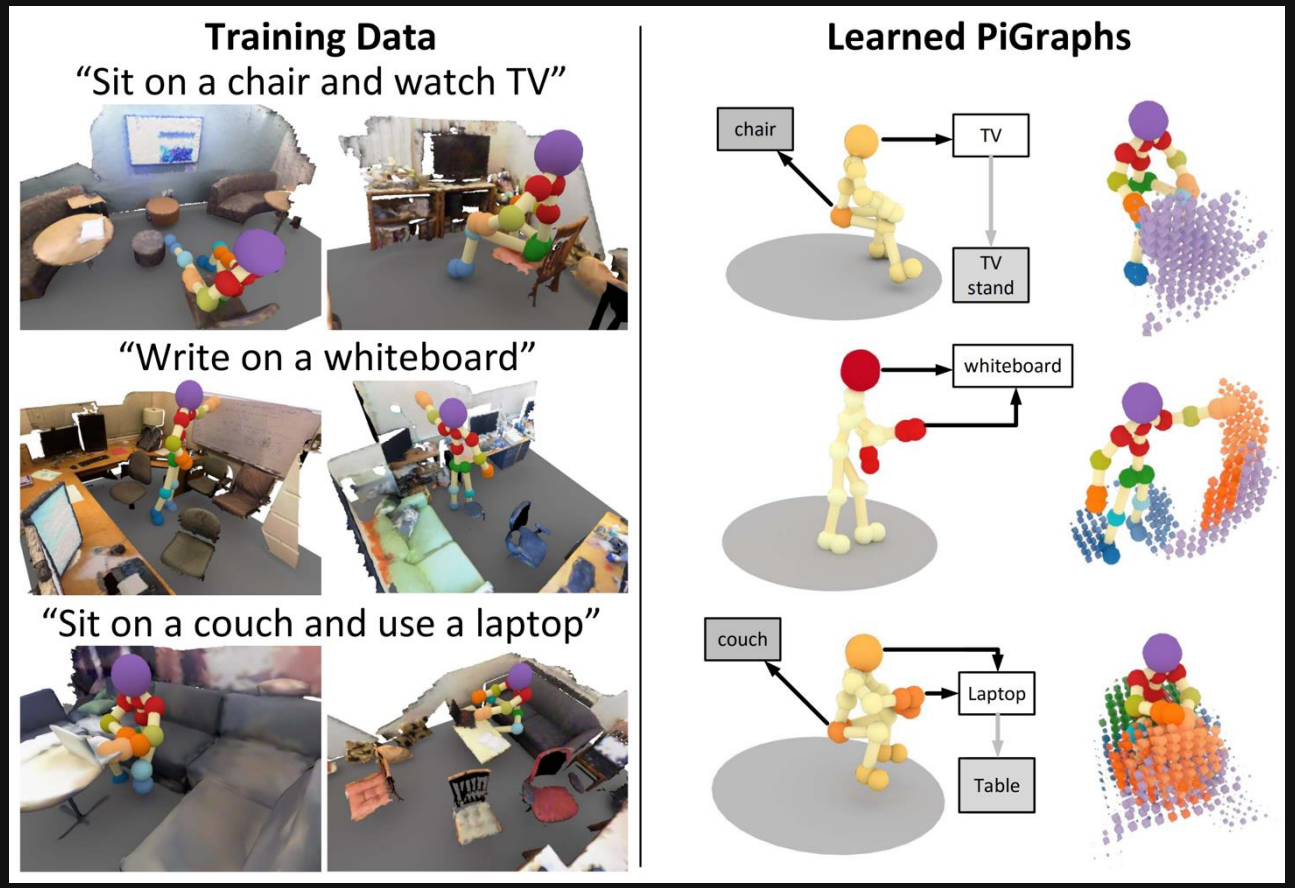


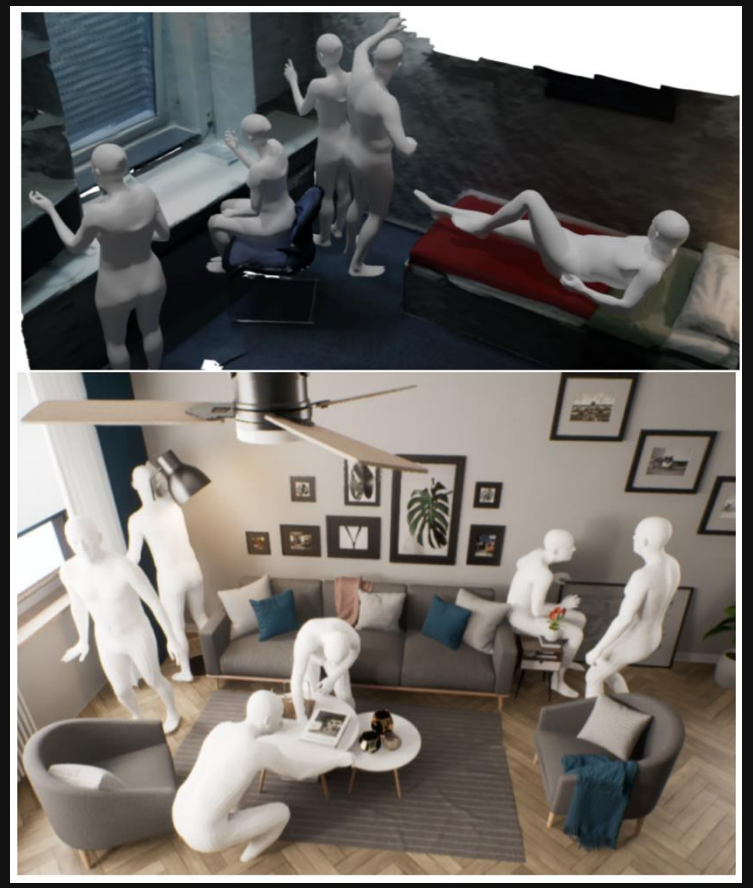
Figure 4: Human-centric representation on a synthetic 3D scene: given an example "sitting" pose, we visualize where a human can sit in the environ-

Put Humans in the Scene

Given a posed 3D body and a 3D scene, can we place the body in the scene so that the pose makes sense in the context of the scene?
That is: does the pose match the affordances of the scene?



PiGraphs



POSA

Typical Tasks in Affordance Learning

- Affordance Detection/Segmentation
- Scene Affordance
- Social Affordance
- …

Social Affordance

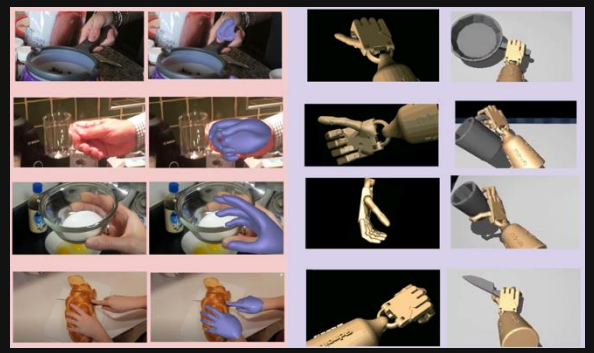
- Social affordance offer possible object-social actions, which may be positive such as sitting on empty chair or negative (socially forbidden) such as opening a lady bag.
- Find an activity recognition mapping $f: (X, E) \rightarrow Y$
  - $X = \{x_1, x_2, \ldots, x_n\}$ denotes the set of object instances
  - $E = \{e_1, e_2, \ldots, e_n\}$ denotes the object interactions for each instance
  - $Y = \{y_1, y_2, \ldots, y_n\}$ denotes the activity labels

Take-aways

- Affordance is all you need when you are using an object/a scene.
- Affordance is NOT the same as object class label.
- Learning object/scene affordances needs a deep understanding of geometry, physics, human poses, activities, etc.
- Understanding affordance is inevitable in the process of building a real robot.

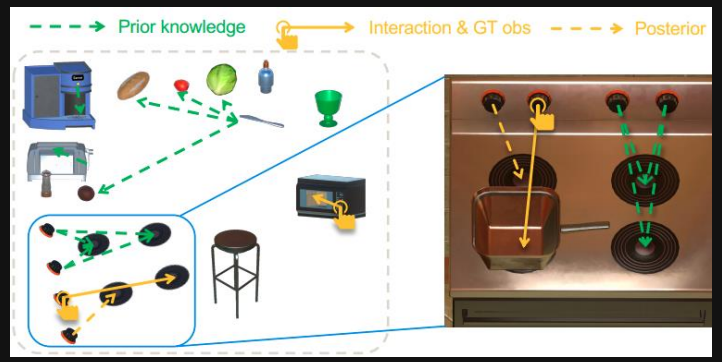
Future Directions

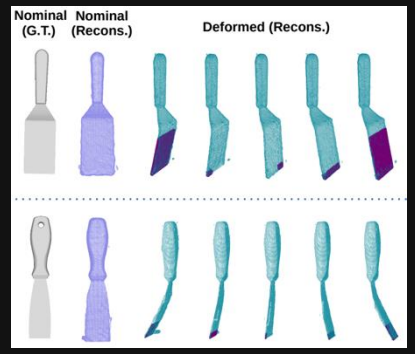
Fine-grained





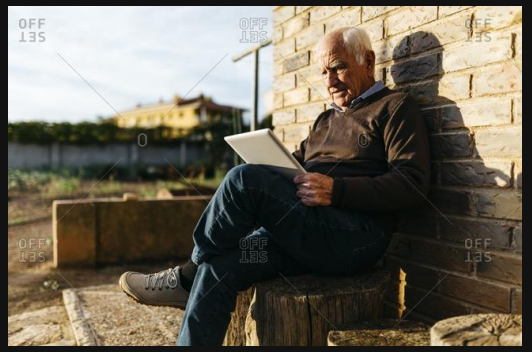
Interaction-rich





Generalizable





Outline

1. Cognitive bases for affordance
2. Computational bases for affordance
3. From affordance to functionality





Human-made objects provide various functions for different tasks


sheep, goats, pigs and cattle. Houses were built and people began to settle. Flint was not found in the Cotswolds. It was mined from parts of southern, western and northern Britain and brought here. People became more skilled at making tools and began to produce new tool types. Fine polished stone axes were used to chop down trees and leaf-shaped arrowheads were used for hunting and weaponry. The first clay pots were made. These were used for storing food, cooking, drinking and ceremonies.
Monuments appeared in the landscape for the first time. Large ditches were dug and banks of earth raised to construct causewayed enclosures. They were the focus for groups of people to come together for important ceremonies. Long barrows were mounds of earth and stone. They contained chambers that housed and commemorated the dead.
Neolithic Stone Tools
01 Flint arrowheads
Including leaf-shaped arrowheads
02 Flint piercers and awls
For making holes in animal skins
03 Flint fabricators
04 Flint knives and serrated blades
Used as cutting tools
05 Hammerstones
Used to strike the flint to make flakes and blades
07 Flint cores
Neolithic Polished Stone Axes



The question ‘What makes things seem alike or different?’ is one so fundamental to psychology that very few psychologists have been naïve enough to ask it.

--- Fred Attneave (1950)



> “If you know what can be done with a ... object, what it can be used for, you can call it whatever you please”
>
> by J. J Gibson

- The problem of understanding the functions aims to identify the tasks that can be performed with an object.

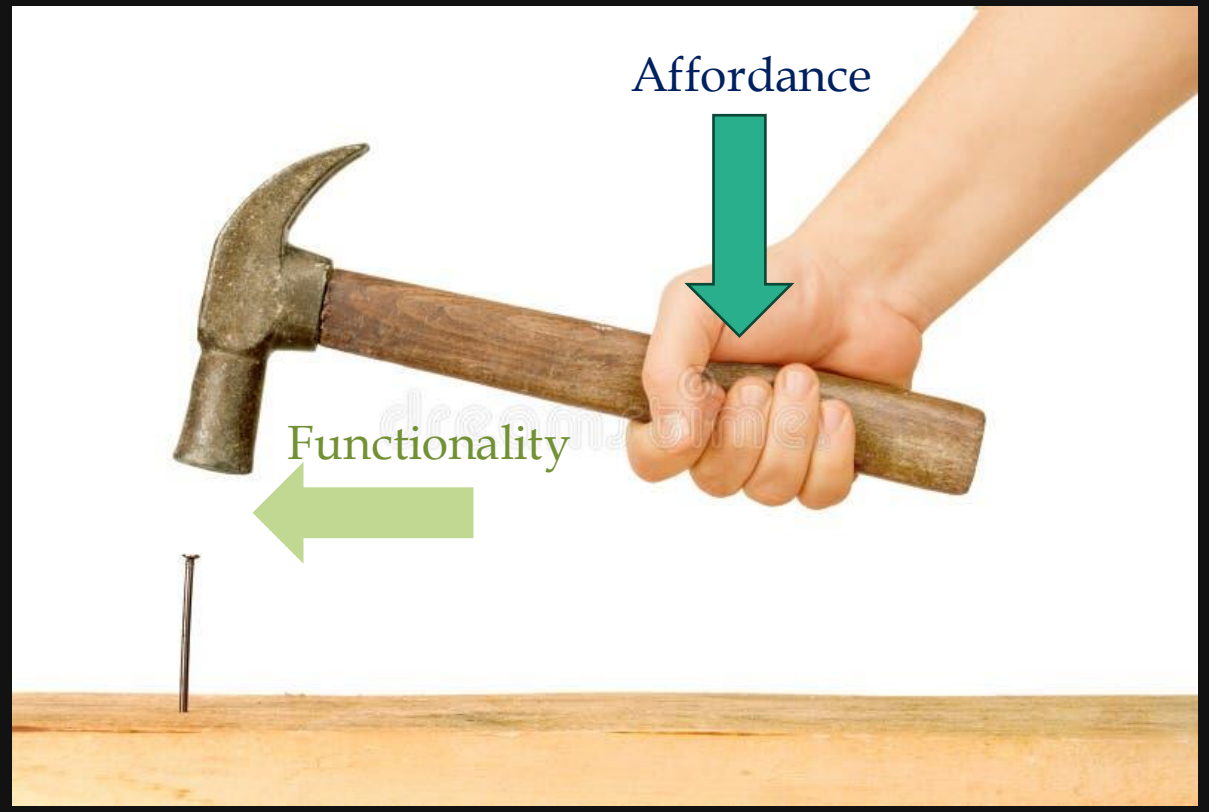
Affordance ≠ Functionality

In contrast to affordances which are directly dependent on the actor, function understanding relates to identifying the possible set of tasks which can be performed with an object. Object function is therefore a permanent property of an object independent of the characteristics of the user.



Tricks to opening a bottle of beer without a bottle opener:
Same affordance, different functionality

Affordance ≠ Functionality



What makes things seem alike or different?



a market that sells kitchenware



a corner in a kitchen

- Have similar objects, e.g., plates, cups, bowls, etc.
- But they offer different functionalities

What makes things seem alike or different?



a modern kitchen



an old-style kitchen

- Have drastically different geometry and appearances
- But they offer similar functionalities

What makes things seem alike or different?



a modern kitchen

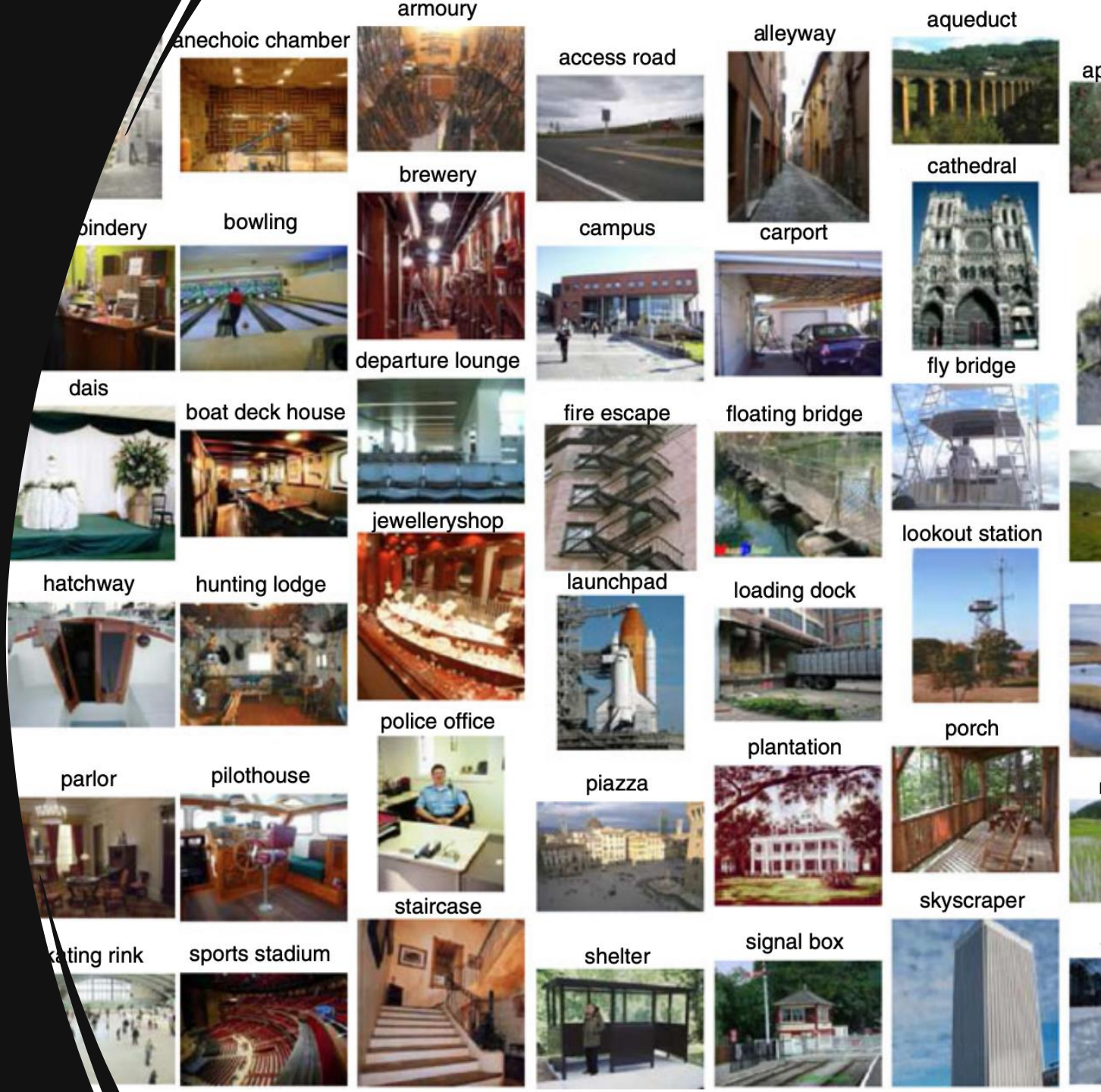


an old-style kitchen

- Scenes with different geometry and appearance could afford the same functionality
- Functionality is a more invariant concept for scene understanding

Functionality understanding of a scene

- Functionality could be recognized from object function-parts



How to infer these tool‘s functionality?



Categorize According to Functionality





- Tool is not a template matching problem, but a cognitive understanding problem.
- We will discuss this topic in later chapters …

Outline

1. Cognitive bases for affordance
2. Computational bases for affordance
3. From affordance to functionality

Thanks for your listening and sleeping